



Renewal Proposal International Research Laboratory **CROSSING**

2026-2031

IRL 2010

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GLOSSARY

AI:	Artificial Intelligence
ANR:	Agence Nationale de la Recherche / French National Research Agency
ANU:	Australian National University
AR:	Augmented Reality
BBB:	Behaviour-Brain-Body Research Centre
CNRS:	Centre National de la Recherche Scientifique / National Centre for Scientific Research
COLREGs:	Collision Regulations
DRL:	Deep Reinforcement Learning
ENSTA:	École Nationale Supérieure de Techniques Avancées / French engineering school
IMTA:	Institut Mines-Télécom Atlantique / French engineering school
IVE:	Interactive and Virtual Environments
LLM:	Large Language Models
LOTUSim:	Naval Group's multi-physics simulation platform for maritime research
MARL:	Multi-Agent Reinforcement Learning
MR:	Mixed Reality
O&M:	Operations and Maintenance
RoboCup:	Robotics competition with various leagues, including RoboCup@Home
SDA:	Space Domain Awareness
UAV:	Unmanned Aerial Vehicle
UI:	User Interface
UUV:	Unmanned Underwater Vehicle
VR:	Virtual Reality

Executive summary

The IRL CROSSING initiative, launched in 2021 and headquartered in Adelaide, Australia, aims to advance collaboration among humans, artificial intelligence, and autonomous systems across sectors such as defense, energy, healthcare, and more. It unites multidisciplinary expertise from CNRS, IMT Atlantique, and leading Australian universities, alongside industry partner Naval Group.

CROSSING 2.0 centers its scientific work on three core research axes —**Autonomy**, **Interaction**, and **Adaptation**— to address critical challenges in **Human-AI teaming**.

Autonomy explores robust systems capable of operating independently while respecting human-defined rules and adapting to dynamic and unpredictable environments.

Interaction investigates seamless communication between humans and autonomous systems, including advanced multimodal interfaces and neurodesign approaches.

Adaptation emphasizes real-time responsiveness to environmental changes and the long-term integration of autonomous agents in human teams.

These scientific priorities are supported by applications in key domains:

- *Maritime*: enhancing collaborative mission and multi-domain operations;
- *Energy*: improving user friendly grid stability and enabling autonomous maintenance of renewable energy systems;
- *Space*: advancing human-robot interaction for long-duration missions in extreme environments;
- *Assistive Robotics*: developing adaptive robots to support humans in unstructured and high-stakes settings.

With the integration of new partners ENSTA (France) and ANU (Australia), CROSSING 2.0 expands its scientific network and geographic reach to Canberra. The inclusion of such institutions in CROSSING 2.0 builds upon an already robust network of core partners, including the University of Adelaide and the University of South Australia (soon to merge as Adelaide University), Flinders University, Naval Group, IMT Atlantique, and CNRS. This expansion not only enhances the initiative's scientific foundation but also fosters deeper collaboration across cutting-edge research domains, strengthening CROSSING's ability to address complex challenges in human-AI teaming.

By aligning French and Australian expertise and leveraging state-of-the-art facilities, CROSSING 2.0 fosters technological leadership, workforce development, and strategic applications, while contributing to the national priorities of both countries and advancing global innovation.

1. Overview

Objective

IRL CROSSING aims to provide solutions for humans, artificial intelligence (AI) and autonomous systems to collaborate together, effectively and ethically.

A strong focus is given to multidisciplinary, to help support certain industries such as environment, health, energy, defense and “industry 4.0”.

Partners

Located in Australia, and launched in early 2021, CROSSING brings together CNRS, IMT Atlantique with three major Australian universities (University of Adelaide, University of South Australia, Flinders University) as well as Naval Group, European leader in naval defense. Below is CROSSING geographic mapping:



Figure 1: Geographic Mapping of CROSSING Partners.

University of Adelaide and University of South Australia are merging as Adelaide University. ENSTA (France) and ANU (Australia) applied to officially join CROSSING 2.0 in 2026. Their proposal to integrate CROSSING 2.0 are in appendix, including scientific inputs and resources contributions.

Below is CROSSING 2.0 geographic mapping:

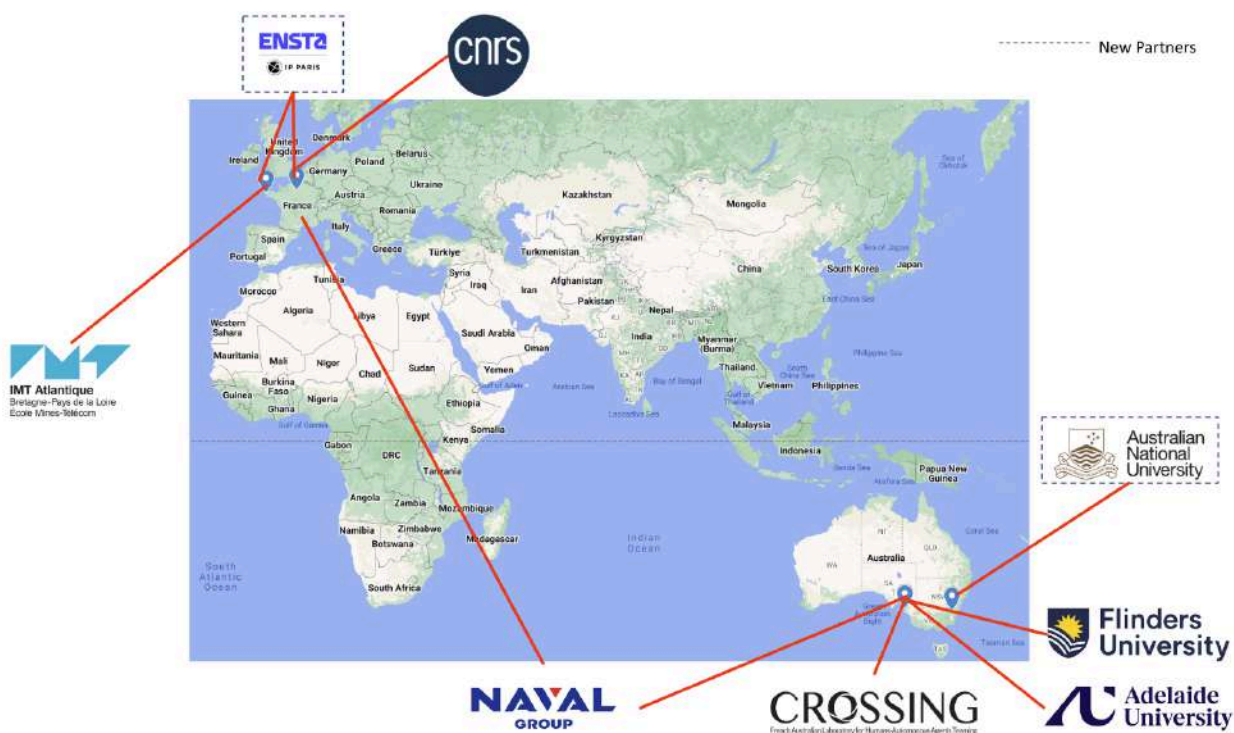


Figure 2: Expansion of CROSSING to Include New Partners.

Despite the time difference between France and Australia, collaboration between scientists is seamless thanks to well-coordinated schedules, digital communication tools, and the flexibility to overlap during early mornings in France and late afternoons in Australia.

Localisation

CROSSING is headquartered in Adelaide. With the launch of CROSSING 2.0, the initiative will expand to include Canberra (ANU) as a satellite location. Following is CROSSING 2.0 geographic mapping, limited to Australia:

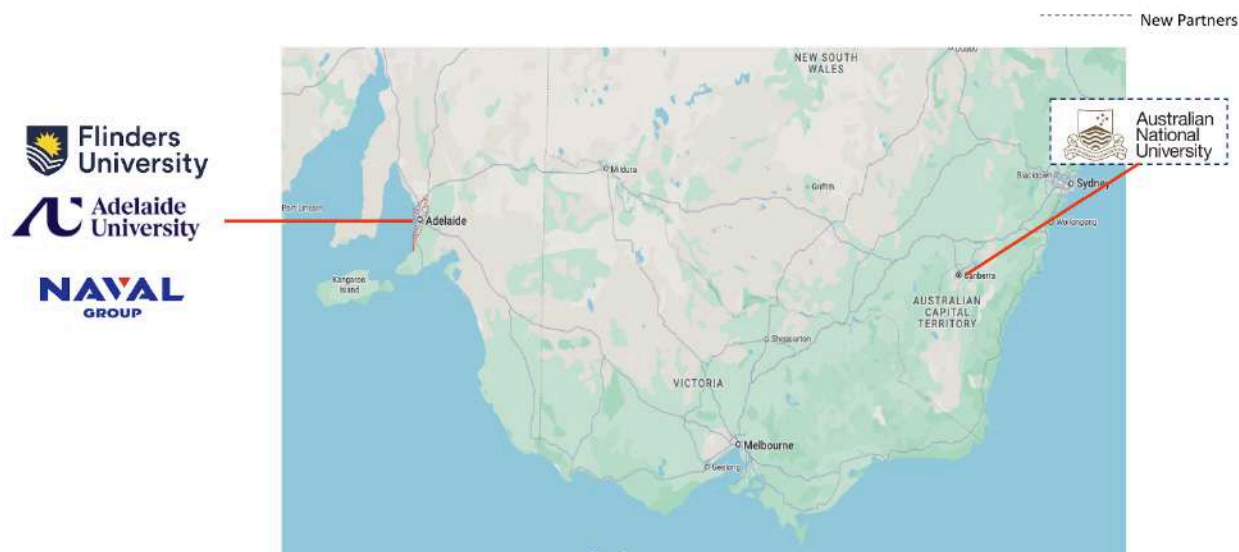


Figure 3: Time Zone Coordination for CROSSING Collaboration.

Despite the slight time difference between Canberra and Adelaide, effective collaboration is easy to manage as the time zones are only 30 minutes apart, allowing for seamless coordination during standard working hours. Despite the 1,150 km distance between Canberra and Adelaide, collaboration is easily facilitated through efficient digital communication tools, regular virtual meetings, and the availability of frequent direct flights, which connect the two cities in just under two hours, ensuring smooth coordination for both daily interactions and occasional in-person meetings.

CNRS Interaction

CNRS CEO Antoine PETIT visited Australia in November 2024, providing an excellent opportunity for Cédric BUCHE, the future CROSSING director, to discuss the IRL CROSSING renewal initiative. Two other IRLs have been launched in Australia, Cédric BUCHE has begun interactions with Sylvie MONNIAUX (Director of the IRL Mathematics in Canberra) and Frederic HOLLANDE (Director of the IRL Medical in Melbourne).



A. PETIT and C. BUCHE



C. BUCHE, S. MONNIAUX and F. HOLLANDE

Renewal Timeline

The renewal process is following a timeline defined with partners :

- Future Director appointment : **Cédric BUCHE** (June 2024)
- ENSTA application to join (July 2024)
- Scientific proposal beta version for review (Nov 2024)
- ANU application to join (Nov 2024)
- Scientific proposal presentation after review (Nov 2024)
- Scientific proposal validation (Jan 2025)
- CNRS visit at Adelaide (April 2025)
- Scientific proposal submitted to CNRS (Spring 2025)
- Agreement submitted for review (Mid 2025)
- Agreement signature (Nov 2025)

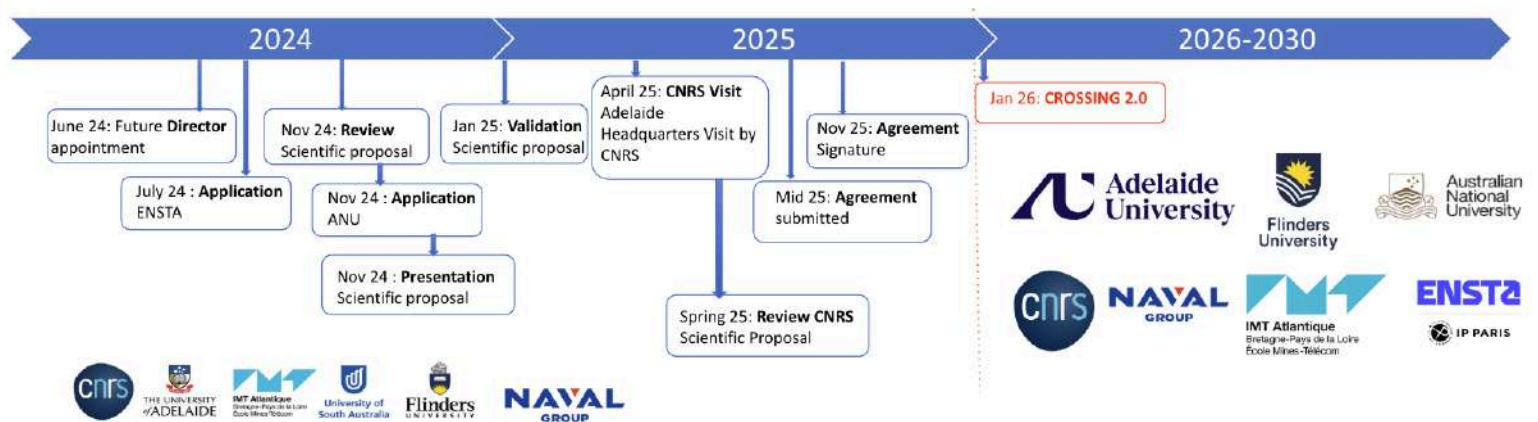


Figure 4: Renewal Timeline for CROSSING 2.0.

2. Scientific Program

Context

Autonomous systems possess the capability to interpret their environment, select appropriate courses of action to achieve objectives, and adapt both their actions and objectives based on evolving situational factors. Recent advances in artificial intelligence (AI), particularly in machine learning and automated decision-making, have facilitated the deployment of these systems in controlled settings. Challenges remain present in the context of a partially unknown environment, especially in complex and heterogeneous systems. When a **human is in the loop**, interacting with autonomous systems, exchanges and situational awareness [Endsley, 1988] becomes key concepts to guarantee safe and effective human-machine interaction. Connected concepts, such as adaptive autonomy and adaptive interaction or co-learning, should be developed to describe the dynamic interactions between an autonomous agent and its human operator [Goodrich & Schultz, 2007]. Within the fields of human factors and human-computer interaction, this adjustable interaction has been framed as Human-Agent Teaming or Human-Autonomy Teaming [Chen & Barnes, 2014]. From an AI perspective, this concept is often referred to as mixed-initiative, in which both humans and machines contribute to decision-making processes in a collaborative manner [Ai-Chang et al., 2004]. The application of these concepts to complex configurations—such as one or multiple human operators interacting with a heterogeneous team of autonomous systems—remains challenging.

IRL CROSSING aims to fill this gap by developing and evaluating interactions between autonomous systems and human operators, particularly when the humans are an integral part of the multi-agent team in real-world and dynamic environments.

The IRL is dedicated to this question and synthesizes different research axes focusing on capacities of adaptation of machines, management of human-system cooperation within augmented environments, as well as development of new ways for intuitive interaction and dialogue between humans and machines.

IRL benefits from its industrial partner, Naval Group, to foster large-scale real-world use cases.

CROSSING 2.0: The Next Chapter

CROSSING's renewal reflects the scientific continuity of its original roadmap, established five years ago. The initial program identified four key themes to address human-teaming challenges: 1/ enhancing the online understanding of humans within hybrid teams; 2/ implementing and embedding learning and generalization algorithms in living environments while exploring their embodiment; 3/ investigating novel interaction paradigms to improve the explainability of autonomous behaviors for human partners; and 4/ developing innovative models and solutions for task and activity management within hybrid teams. These foundational themes have steered CROSSING's research and remain central to the expanded ambitions of CROSSING 2.0.

The pandemic significantly disrupted the early stages of CROSSING, curtailing international exchanges and in-person collaboration. Despite these challenges, the initiative demonstrated remarkable resilience, engaging in 24 PhD initiatives, securing the world championship of social robots twice in a row (RoboCup@Home), and earning support through the Franco-Australian FACET funding program.

Geopolitical events, such as the cancellation of the submarine contract between Naval Group and the Australian government, introduced further uncertainty. This setback delayed funding mechanisms and affected early collaboration dynamics. Nevertheless, CROSSING rebounded strongly, achieving milestones such as 29 co-authored publications and launching new research projects with French and Australian institutions. Notably, Naval Group's renewed engagement has provided CROSSING with two engineers, a full professor, and funding for three PhD projects in the last year.

CROSSING 2.0 aims to capitalize on this progress while expanding its scope. The inclusion of new partners, such as the Australian National University (ANU) and ENSTA, brings additional expertise to the consortium. Furthermore, the establishment of Canberra as a satellite location enhances CROSSING's geographic footprint and visibility, fostering dynamic collaboration between Australian and French researchers. Additionally, colleagues from IRP-ARS (University of Melbourne, ANU) and former colleagues from the University of Adelaide (now at Monash University) are committed to CROSSING 2.0.

These expansions align seamlessly with CROSSING's scientific goals, enabling deeper exploration of its core research themes: Autonomy, Interaction, and Adaptation (see next sections). By promoting multidisciplinary research and connecting diverse expertise, CROSSING 2.0 ensures continuity in its scientific mission while adapting to emerging challenges.

Lessons learned during CROSSING's initial phase have informed a more resilient operational structure for CROSSING 2.0. Recent achievements, such as the recruitment of a dedicated project manager, have strengthened administrative and strategic support. Additionally, new workshops focusing on ethical AI and human-robot teaming have laid the groundwork for impactful collaborations and groundbreaking research initiatives.

With the ongoing support of its core partners and the integration of expanded resources and expertise, CROSSING 2.0 is well-positioned to achieve significant advancements in Human-AI teaming. This renewal not only builds on the foundation established in its first phase but also solidifies CROSSING's role as a global leader in addressing complex challenges through innovative research and international collaboration.

Challenges

For nearly four years, IRL CROSSING has been investigating how humans and AI systems can collaborate effectively in multidisciplinary research programs. To address this challenge, Australian and French researchers have proposed a new structure for CROSSING 2.0, centered around three core pillars that seamlessly integrate both theoretical and practical approaches:

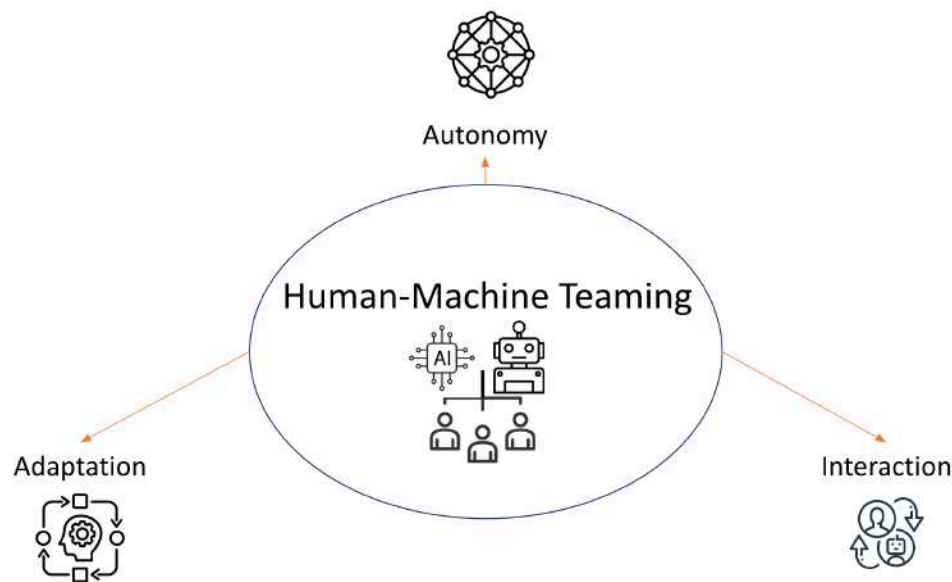


Figure 5: Research Axes in CROSSING 2.0: Autonomy, Interaction, and Adaptation.

1. Autonomy — Axis 1

Keywords: Autonomous Systems, Complex System, Human in/on the loop, Control, Multi-agent

2. Interaction — Axis 2

Keywords: Situation Awareness, Common Sense, Dialogue, Interface, Event Recognition, Neurodesign

3. Adaptation — Axis 3

Keywords: Dynamic and Unknown Environments, Adaptive Autonomy, Trust, Fatigue, Attention, Co-learning, Adaptive Interface, Decision Support System, Cognitive Load, Human-modeling

Axis 1: Autonomy

Key Idea:

Autonomous behaviors generally refer to pre-defined high-level objectives, which have been planned by human operators or authority, and which define the technical aspects and level of performance that are expected from the system (or set of systems). The autonomy will be revealed in the system's ability to tackle undefined or ill-defined low level events (e.g. currents influencing underwater trajectories) or in the way the system will handle unexpected and unplanned events on the duration of its "autonomous" mission. By definition, this ability will result in the systems achieving the requested final objective, but through multiple and unpredicted / unpredictable options or fine grain behaviors modalities. This is further complicated in the context of the autonomous systems teaming, where agents act both individually and collectively.

CROSSING 2.0 aspires to address complementary aspects of this conventional approach, whilst dealing with the following topics:

- **Human in/on the loop:** CROSSING 2.0 is mostly focusing on socio-technical systems that include not only smart and autonomous systems but also human operators. From a general point of view, these systems operate according to rules that have been set by humans, usually for the sake of security or the compliance or the systems modes with social constraints and norms. Taking this specificity into account, one of the research objectives would be to develop algorithms and systems that show respect to the human rules while simultaneously optimizing their self and local behavior [Simon et al, 2023]. This should lead to the development of "hybrid AI models" extending logical and physical rules in the core of the learning process. These models would extend beyond standard machine learning approaches by integrating human values/rules into their core, making them more responsive to socio-technical dynamics.
- **Human in the group:** Systems have recently and rapidly evolved from centralized towards distributed nature, raising multiple challenges. The autonomy (reaching a high-level goal as defined by operators or authorities) is now to be supported by no more unique but collective behaviors, and the success may be achieved through an emerging macroscopic behavior based on agentic local elementary ones. As simple examples, one could mention a successful mission for a set of attacking agents if one of them amongst a large set reaches a target, or another one that reaches the objective while covering collectively an area to be patrolled. The objective here will be to propose to model and evaluate multi-agentic-AI systems (including potentially heterogeneous agents abilities) according to this paradigm [Hadi et al., 2024; Gautier et al., 2023]. Of course, one or several agents within the set could be humans, so that their own and specific capacity to contribute to the collective goal should be studied. Considering the previous topic could also be meaningful while putting one of the human agents in charge of defining, communicating and controlling high-level socio-technical rules amongst the multi-AI-agentic system itself.

- **Taking into account human limitations and capacities:** From a pure perception and cognitive point of view, some special constraint issued from the human agent(s) within the collective can also lead to restriction in the richness and variety of behaviors of the multi-agent system. Beyond the socio-technical, logical or physical rules mentioned here above, these special constraints - expressing the “human limitations” - and their impact on the systems’ performance will be studied [Simon et al., 2023; Garcia Canderas & Tapus 2024]. The understandability of the systems’ behavior could be especially impacted by this approach.
- **Control in complex situations:** Control in human-machine teaming is essential for effective collaboration between humans and autonomous systems, particularly in complex, dynamic, and high-stakes environments. Traditional control methods fail to handle the full range of operating conditions in complex dynamical systems. Recent learning-based adaptive control methods design systems to compensate for unknown dynamics and non-linearities by combining control theory with deep learning or advanced data-based techniques [Iannelli & Postoyan, 2024]. While stability issues with neural networks have been addressed, machine learning can now extract underlying dynamics, enabling more flexible and robust systems [Chaffre et al., 2024]. Research aims to define task-specific models, learn dynamics from system feedback, and develop model-predictive methods to address uncertainties. A key challenge is ensuring safety, particularly stability [Granzotto et al., 2024], in high-stakes applications (with humans involved) where failure is unacceptable.
- **Bio-Inspired Learning:** Current machine learning systems are trained solely for the objective of minimizing loss functions. Traditionally, a large amount of data is fed under an independent and identically distributed fashion to the learning systems. Despite being ideal from an optimization point of view, there is evidence [Hayes et al., 2021] showing that this procedure is far from being optimal from a pure learning perspective where the goal is to maximize the measure of utility of the system while minimizing the overall process costs of the learning process. In nature, learning systems tend to prioritize learning efficiency as much as accuracy, which in the long run seems to lead to higher generalization abilities. Some prior works [Chaffre et al., 2023] aimed at incorporating these missing biological elements in our machine learning systems, but a research objective would be to develop novel solution methods that simultaneously take into account these dual objectives.
- **Multi-Agent Reinforcement Learning:** Deep Reinforcement Learning (DRL) with multi-agent systems is fundamental to human-machine teaming, empowering autonomous systems to learn, adapt, and collaborate fluidly in dynamic and complex environments. In such settings, human operators and machines must work in close synergy, merging individual expertise with the collective intelligence of multiple agents to address challenges that neither could solve in isolation. Multi-agent systems are particularly suited when individuals face constraints such as limited endurance, range, or effectiveness when operating alone. By leveraging information fusion and resource complementarity, multi-agent systems can accomplish missions that are beyond the capabilities of any single agent [Pang et al., 2023]. In unpredictable environments, the ability to adapt is crucial, and Multi-Agent Reinforcement Learning (MARL) has become

a key tool to enable systems to learn in real-time, adjusting their behavior to integrate human feedback and respond to dynamic situations [Busoniu et al., 2008]. Future challenges will include enhancing the scalability of MARL systems to handle larger, more complex teams, ensuring reliable coordination and communication among agents in uncertain environments, and establishing robust safety mechanisms to prevent system failures in high-stakes applications.

- **Advancing Autonomy for Uncrewed Robotic Systems:** Human-machine teaming is key to advancing uncrewed systems, where autonomous machines collaborate with human operators to tackle complex tasks requiring both human intuition and machine precision. This research aims to enhance individual and networked uncrewed systems. While the technology to build these machines exists, challenges remain in ensuring robust perception algorithms to interpret complex environments in real time, and control algorithms to guide the system safely through this complexity. Additionally, these systems often have poorly understood dynamic models and noisy sensors, requiring operation near vehicle performance limits and in dynamic, unknown environments. The goal is to advance systems and control theory to improve the robustness and reliability of uncrewed systems in complex real-world settings. The progress made during the previous IRP-ARS period provides a solid foundation for the next research phase, focusing on leveraging non-linear and symmetry-based insights to combine high-performance control with reliable real-time state estimation and situation awareness, addressing high-impact open questions
- **Dynamic Hierarchy Management in Multi-Agent Autonomous Systems:** Hierarchical structures, typically defined by human operators, may become rigid or inefficient when unexpected events or evolving mission dynamics demand rapid adjustments. Autonomous systems must dynamically reconfigure their roles and interactions, balancing individual initiative with collective goals [Paul et al., 2021]. This requires innovative approaches to flexible decision-making frameworks, enabling agents to reassess and adapt their hierarchical relationships in real-time while ensuring alignment with the overall mission objectives.

Outcome:

The outcome of this research is a comprehensive framework for advancing autonomy in socio-technical systems, integrating human-centric rules with hybrid AI models to optimize both individual and collective agent behaviors. By addressing human cognitive limitations and leveraging bio-inspired learning, the work ensures robust, adaptive, and efficient multi-agent systems capable of operating in complex, dynamic environments. Novel methods in learning-based adaptive control and reinforcement learning enable these systems to handle uncertainties and enhance collaboration, aligning technological advances with ethical and operational requirements. The development includes evaluating agent behavior in terms of compliance with a set of rules, particularly human-based ones, across varying levels of human involvement. This effort establishes foundational benchmarks and tools for future development of reliable, human-compatible uncrewed robotic systems in real-world applications.

On-going projects¹

Date	Type	Staff	Partners	Description
2024-2028	PhD	Ignacio Meza	Adelaide Uni / Naval Group	Ship classification in complex maritime context
2024-2028	PhD	Antoine Vivien	Flinders Uni / Naval Group	Robust multi-drone coverage planning
2024-2027	PhD	Ardianto Wibowo	Flinders Uni / IMTA	Qualitative case-based reasoning for swarm of drones navigation
2024-2027	PhD	Lucas Fernandes Marchand	Adelaide Uni / IMTA	Distributed learning on connected devices with limited resources
2023-2026	PhD	Antoine Wanctin	Flinders Uni / ENSTA	Machine Learning for COLREGs
2025-2029	ARC	Dragan Nesic, Romain Postoyan	University of Melbourne	Optimisation-based control for multi-agent systems
2024-2028	PhD	Robert Moldenhauer	University of Melbourne, Université de Lorraine	Stability guarantees for stochastic (near-)optimal control problems
2024-2027	PhD	Jonathan de Brusse	University de Lorraine / University of Melbourne	Dynamic programming and discounted costs: near-optimality and stability guarantees
2024-2028	ANR	Romain Postoyan, Jonathan de Brusse	Université de Lorraine / CNRS	Control by neuro-dynamic programming: stability, robustness and optimality
2025-2028	PhD	Samy Ait Touati	University Côte d'Azur / University of Melbourne	Decentralized estimation and distributed control for a formation of unmanned surface and underwater vehicles

*Table 1: Current PhD and Postdoctoral Projects in Autonomy.***Futures actions :**

- CNRS delegation / affectation at CROSSING 2.0:
 - Tarek Hamel (2027-2029)
 - Benoit Clement (2029-2030)
 - Lucia Bergantin (2028-2029)
 - Christophe Viel (2028-2029)
- PhD Proposal 2025-2028 :
 - Collaborative mission (Naval Group)

¹ In this document, we consider only projects that will continue beyond 2026

Axis 2: Interaction

Key Idea:

To enhance collaboration in hybrid teams of humans and AI/Autonomous Systems, this research explores interaction through advanced solutions for language and nonverbal communication strategies, integrating methods that address both approaches cohesively.

A first central topic for interaction addresses the language and modalities to be used within hybrid teams of humans and AI /Autonomous Systems. The recent explosion of Large Language Models (LLM) has potentially hidden some of the most challenging aspects of efficient and adapted models able to handle dialogs between humans and AI / Autonomous systems. CROSSING 2.0 would like to explore the use of advanced logical and dialogical tools to enrich, complement or qualify the limits of “usual” LLM capabilities. This topic could be addressing challenges such as:

- **Maintaining common sense in Human-AI dialogs:** A remaining crucial question of LLM is related to their capacity of expression, their ability in respecting and efficiently using underlying structures tied to language and creating sense for humans. As a simple example, an AI should be able to use qualitative vocabulary and speak about “far, close, very close” or “small, average, big” while monitoring the quantitative uncertainty attached to the use of those words [Egré et al., 2023]. This involves reflecting on how humans themselves learn and negotiate the vague meanings of such words, and an ability to determine for qualitative vocabulary both typical values and most likely interpretations as a function of the context. This research should lead to the study of the ability of current LLM models to express such semantic fields, potentially propose some new models, and especially hybrid models of learning under semantic constraints [Icard & Égré, 2023]. It should also include the study of human characteristics and “needs” in terms of common sense in order not to overestimate and misadjust the dialogue requirements (this could of course be task and domain dependent, to be discussed). With the fast development of LLMs, related work needs also to be conducted on both the way lexical representations in AI and in humans are deployed, and on how logical words and inference are linked in LLMs.
- **Subtly lying:** Rich dialogs do not simply tell everything every time, and do not always tell the truth, for bad or good reasons [Egré & Icard, 2018]. If we consider bad reasons, one could address the ability of AI and autonomous systems to deceive opponents. This could be a result of models associating the understanding of opponents’ understanding of a situation, as well as their intentions, and a tactical / strategic way of displaying our own intention and state. Research on communication, explicability, lying by omission and deceiving could be addressed in this way, and human AI teaming could be considered as a stronger configuration to reach this objective, building the specific human autonomy cooperation to be able jointly subtly lying to the “external interlocutors”. As for the good reason, the objectives may rely on the same logical and mathematical tools and models but are completely different. In this case, the autonomous / intelligent system would rely on the lying (mostly by omission) to facilitate the reaching of a common ground and good

understanding from the human operators' side. As a simplistic example, one could think about an AI omitting the divergent pieces of a situation description to insist and emphasize its central and main structure, to rapidly set a common ground and build decision. Time and modalities of information communication would also be addressed, as dialog must be considered here as a process and not a one-shot information sending and receiving. The use of nudges by the system could also be considered..

A second topic of research on interaction addresses the use of nonverbal communications and especially the impact of immersive displays and adequate metaphors to express and represent the complexity emerging from the AI / Autonomous systems / Sets of Autonomous systems. Augmented reality and mixed systems will be the support for such a topic, but its main target should consist in exploring the best ways, modalities, shapes, metaphors to represent the autonomy / Ai as a partner.

- **Metaphors and representations for intelligence**: One of the questions to be addressed will be the impact of the form to be taken by the intelligent agents to dialog and cooperate with, while considering the complexity and new modalities of communication and dialog mentioned here above. The impact on acceptability of avatars vs. synthetic agents [Even et al., 2021], unique entity vs. distributed representation of collective intelligence, stable vs. adaptive context-based representation of intelligent agents will be studied. This approach should be highly task dependent, and one of the objectives will be to propose a general methodology allowing to pick up and adapt the most adequate metaphor from the context.
- **Agentic visualizations**: Matching the agentic nature of autonomous systems as described in the autonomy axis, one of the most challenging aspects of the immersive interactions will consist in being able to express this diversity without saturating or overwhelming the operators in the system based on their current cognitive capacity and focus [Rinnert et al, 2023]. This research axis will certainly rely on the main trends in data visualization, but the specificity of dealing with dynamic and “smart” data as supported by a dedicated autonomy.
- **Mixed reality and robotics**: Integrating mixed reality with robotics explores how these technologies can create a cohesive environment for both human operators and autonomous agents. This will focus on developing Mixed Reality (MR) interfaces that allow humans to interact with robotic systems as if they were extensions of their own operational space, creating fluid, intuitive interfaces that facilitate collaboration and improve the overall effectiveness of the systems [Buche et al., 2023].
- **A Spatial Common Ground**: MR is envisioned as a collaborative spatial environment that provides a platform for real-time interaction and adaptive teaming between humans and AI agents. Rather than relying solely on Virtual Reality (VR), MR creates a Spatial Common Ground, an immersive shared space where humans and autonomous systems can engage in seamless cooperation. In dynamic or uncertain scenarios, MR allows for intuitive adjustments and mutual understanding, supporting the objective to foster adaptive autonomy and real-time co-learning. This framework visualizes complex and evolving team dynamics, enabling effective decision-making and interaction while

maintaining situational awareness, thereby reinforcing human-AI collaboration in adaptive contexts.

Finally, research including both verbal and non-verbal will be tackled:

- **Human / AI collaboration:** Collaboration between AI and humans is crucial. First, humans bring unique skills, expertise, intuition, experience, and creativity, particularly for building multi-agent autonomous systems. Second, involving humans in the AI process ensures that the system's outputs align with human objectives. Lastly, integrating humans in the AI loop fosters trust and transparency in AI decisions. This research focuses on studying Human-AI collaboration in the construction of multi-agent AI systems and decision-making processes in real time. To tackle this challenge, we cannot merely critique the AI outputs as with chatGPT. Instead, we need a more generic architecture that can decompose complex problems or situations into simpler parts, handle human feedback, manage consensus between multiple human and AI feedbacks, store and retrieve structured knowledge in long-term memory, and combine AI with deterministic algorithms (tools). Human-AI collaboration is an active research topic, and there are some works, such as [Daull et al., 2023], that we can rely on to address our research challenges. This goes beyond efforts in explainable AI and into the multi-modal dialog between system outputs, AI interactions, and the human's own mental model.
- **Applying Neurodesign to the Creation of Sophisticated Interactive Systems:** This stream of research lies at the intersection of human-technology interaction, human-machine teaming, as well as both human and machine prediction and decision-making [Gracia Cardenas & Tapus, 2024]. Here, we will explore mechanisms of human perception and learning and how these cognitive processes can inform the development of sophisticated interactive systems. We refer to this approach as 'neurodesign,' a methodology that our team has previously applied to the creation of mixed digital and physical experiences that incorporate technologies such as virtual reality, augmented reality, and artificial intelligence. Additionally, this stream of research will be exploring the implications of these advancements within speculative and applied futures, aiming to contribute to the ethical and purposeful integration of technology in human environments.

Outcome:

The research aims to revolutionize Human-AI collaboration by developing adaptive, multimodal interaction systems that seamlessly integrate verbal and non-verbal communication, enhance trust, and address human cognitive needs. It explores creating AI models capable of structured, context-sensitive dialogs, utilizing mixed-reality frameworks and metaphoric visualizations to foster intuitive interfaces. By applying neurodesign principles, the work aligns advanced interactive systems with human cognition and ethical considerations, enabling dynamic, real-time co-learning and decision-making in complex environments. This approach ensures meaningful, effective collaboration between humans and AI in hybrid, adaptive teams.

On-going projects

Date	Type	Staff	Partners	Description
2024-2028	PhD	Shanmugaraj Madasamy	Flinders Uni / Naval Group	Event detection from multiple marine drones
2023-2026	PhD	Gaëlic Bechu	Adelaide Uni / IMTA	Combine LLM with object detection to Improve situation understanding
2023-2026	PhD	Hailu Beshada Balcha	UBS / CNRS	Human Robot Interaction Assessment Methods
2024-2028	PhD	Kingsley Stevens	Adelaide Uni / Naval Group	Human-machine immersive interface for tactical situations in maritime context
2025-2028	PhD	Taha Lamine	Adelaide Uni / ENIB / Naval Group	Cross-reality between human and humanoid robot
2025-2028	PhD	TBD (aufrande)	Adelaide Uni / IMTA	Augmented reality and user perception in motion
2024-2027	PhD	Nazial Md	Adelaide Uni / IMTA	AI "perceptive assistance" during tele-operations of rovers in Deep Space

Table 2: Current PhD and Postdoctoral Projects in Interaction.

Futures actions :

- CNRS delegation / affectation at CROSSING 2.0:
 - Paul Egré (2026)
 - Emmanuel Chelma (2026)
- PhD Proposal 2025-2028 :
 - "Vague, Deceptive, Ambiguous COLREGs" (CNRS/Flinders/ENSTA)
 - "AI Deception within Multi-Agent Systems" (Flinders)
 - "Lexical representation in language models" (IRISA-CROSSING)
 - "Logical connectives and logical inferences in language models" (LSCP Paris-CROSSING)

Axis 3: Adaptation

During interaction, humans and machines need to adapt. Adaptation has to be considered here according to two points of view, strongly related to time scales to be taken into account.

From a short-term perspective, within the context of human-machine teaming, adaptation will focus on managing uncertainty and unpredictability of situations. For autonomous systems to be effective teammates, they must be able to learn and adapt in real-time to dynamic and partially unknown environments, taking into account human singularities. These systems must integrate human feedback and adjust their behavior to accommodate uncertainty and environmental changes. CROSSING 2.0 would like to address the following topics:

- Mimicking resilient human behaviors:** This research focuses on adaptive decision-making solutions inspired by human-behavior for the control of robots or AI. The method is based on the idea of mental simulation that aims to evaluate multiple alternatives of decisions before the decision is made [Polceanu & Buche, 2016]. Such an approach means the modeling of the environment and other actors (humans or AI), it also means the possibility to anticipate changes in the environment or in the behavior of others. Different steps in this process progressed with an increasing level of complexity including interactions with one and multiple virtual agents and humans. An architecture by which the three modes – reactivity, predictability, and adaptability – can function asynchronously in parallel. The prediction is made within an autonomous world (simulation within a simulation), in which the entity/robot can physically simulate itself (with its own behavioral model) and its environment (with the representations that it constructs of the behaviors of other entities). This topic is connected to the Digital Twin concept and refers to embodied intelligence frameworks such as GIBSON, AI2THOR and ProcTHOR.
- Strengthening the what-if protocol:** Artificial systems have the ability to generate a very large number of alternative situations and therefore can efficiently proceed with counterfactual reasoning or counter-factual cases generation [Buche et al., 2016]. The topic of this research theme should focus on the use of causality within this field so as to generate more efficient and understandable proposals from the system point of view, as to adapt to the unexpected event.
- Adaptive Autonomy:** CROSSING 2.0 will work on solutions based upon adaptive autonomy levels, whereby the system detects loss of reliability of these models and responds by temporarily lowering its own autonomy levels and increasing engagement of the human in the decision-making process. Shared control mechanisms will foster intuitive and efficient exchanges between human operators and autonomous systems, allowing the system to learn progressively from these interactions. This learning will enhance the system's ability to communicate intent, provide real-time feedback, and maintain transparency in decision-making, ultimately building a foundation of trust. The system will autonomously adjust its interaction strategies based on user preferences and contextual demands while preserving consistency and reliability. This adaptability will be crucial for smoother human-system cooperation, especially in dynamic tasks that require varied levels of autonomy and complex decision-making. In the context of maritime situations, where swarms of underwater drones are deployed for missions with operators, solutions will estimate drones' reliability to adjust the drones' autonomy, from fully autonomous to fully controlled by humans.

- **Adaptive Interface**: CROSSING 2.0 will explore enhancing human-agent collaboration in mission-critical environments by leveraging adaptive user interfaces (UIs) and LLMs to improve trust, communication, and transparency. A key aspect of this research will be examining how AI-driven UIs can dynamically respond to human cognitive load, intent, and experience, facilitating more seamless interactions in high-risk scenarios. By modeling user behavior and predicting operator needs in real time, CROSSING 2.0 aims to enable tailored interaction modalities — such as voice commands, haptic feedback, and augmented reality (AR) [Rinnert et al., 2023] — that adapt to the specific context of the task.
- **Safe autonomous navigation for heterogeneous teams**: In contexts where human operators and autonomous mobile agents work in close proximity to accomplish shared tasks, it is critical for autonomous agents to predict human behaviors and autonomously avoid hazardous situations while navigating unpredictable, unstructured, and partially or completely unknown environments [Clement et al., 2024]. Autonomous agents should prioritize the safety of their human teammates while ensuring their own safe navigation through the surrounding environment (which may also include other dynamic obstacles), in real time. Heterogeneous teams may also consist of multiple human operators and different types of autonomous agents with various degrees of priority, depending on the specific task.
- **System of systems self-adaptation**: Predict the evolution of the system (demand, production) and adaptation to anticipate the changes (eg. switch off parts, use battery farm or cars, real-time pricing, notification to consumers).

From a long-term perspective, the focus will be dealing with the adaptation of individuals and teams to the embedding of autonomous and AI agents within the group. This should take into account some aspects more focused upon psychological and sociological evolution of hybrid systems, as the embedding of AI in operational systems will lead to changes in roles, authority sharing and potentially commitment. This could develop the following research topics:

- **Long-term trust evolution**: Empirical and theoretical models of evolution of trust over long periods should be studied, including the sequences of events, errors, decisions that will have been encountered by the hybrid system [Simon et al., 2024]. This should lead to a way to calibrate the trust (avoiding over and under trust in the system), as well as dynamical mechanisms of task and authority allocation.
- **Management of affect, commitment, responsibility**: Introducing, in a sustainable way, a new member in a community or a team leads to potentially deep changes in the psychology of the other members of the team and could modify their version of self-responsibility, commitment or even could develop some affect-related relationships with artificial companions [Simon et al., 2023]. This could result in profound changes in the socio-psychological balances in the group. This topic aims at studying this kind of evolution and at proposing some methodology and tools to better manage the impact of such changes.
- **Agile governance and policy design for Autonomous Agent/Human Interaction**: This stream aims to address unresolved tensions in governing and policy environments

where human-machine teaming is increasingly prevalent, including defense. It will explore challenges and opportunities in algorithmic and human governance in fast-changing environments (due to technological advances, new research, and evolving environmental, social, and behavioral conditions). This includes developing analytics for assessing risks and opportunities, as well as skills for leading change and managing hybrid teams in complex settings. Building on the state of the art in algorithmic decision theory, policy design, agile governance, and complexity leadership, this work aims to inform governing structures, rules, policies, and operational systems necessary for navigating complex environments.

Outcome:

The outcome of this research is the development of adaptive systems capable of real-time decision-making and seamless integration in dynamic, uncertain environments, where autonomous agents can learn and adjust based on human feedback. It explores models for mimicking resilient human behaviors, enhancing trust through adaptive autonomy and interfaces, and ensuring safe, efficient collaboration between human operators and heterogeneous autonomous agents. Long-term, the research will focus on understanding the evolution of trust, commitment, and responsibility in hybrid teams, and designing governance frameworks that enable effective management of AI-human collaboration in complex, changing environments.

On-going projects

Date	Type	Staff	Partners	Description
2023-2027	PhD	Melissa Ngo	Adelaide Uni / UBS	Fatigue and multitasking in Human machine teaming
2023-2026	PhD	Juan Jose Garcia Cardenas	ENSTA	Performance, Trust, Workload, and Social Processes in a Human-Robot Task
2025-2028	PhD	TBD (FACET)	Adelaide Uni / Grenoble Uni	TBD
TBD	Post doc	TBD (FACET)	Adelaide Uni / Grenoble Uni	TBD
2022-2026	PhD	Sridath Tula	LAMIH / CNRS	Human-Robot Teaming

Table 3: Current PhD and Postdoctoral Projects in Adaptation.

Futures actions :

- CNRS delegation / affectation at CROSSING 2.0:
 - Vincent Debusschere (2026)
 - Gilles Le Chenadec (2029-2030)
- European MSCA, “HyCI” for Hybrid Collective Intelligence, J. Diguët, P. Egge, MQ Pham, E.Abbasnejad, J. Chi

Use Cases

Previous scientific priorities — Autonomy, Interaction and Adaptation — are supported by applications in key domains: Maritime, Space, Energy and Assistive Robotics. Align with scientific foundations, these domains were selected based on the expertise of our partners, ongoing funding initiatives (e.g. FACET), state-of-the-art facilities (e.g. Andy Space at Adelaide), and the success of CROSSING 1.0 (e.g. RoboCup@Home competition). These elements provide a strong foundation for advancing research and innovation, leveraging available resources and expertise to drive meaningful applications.

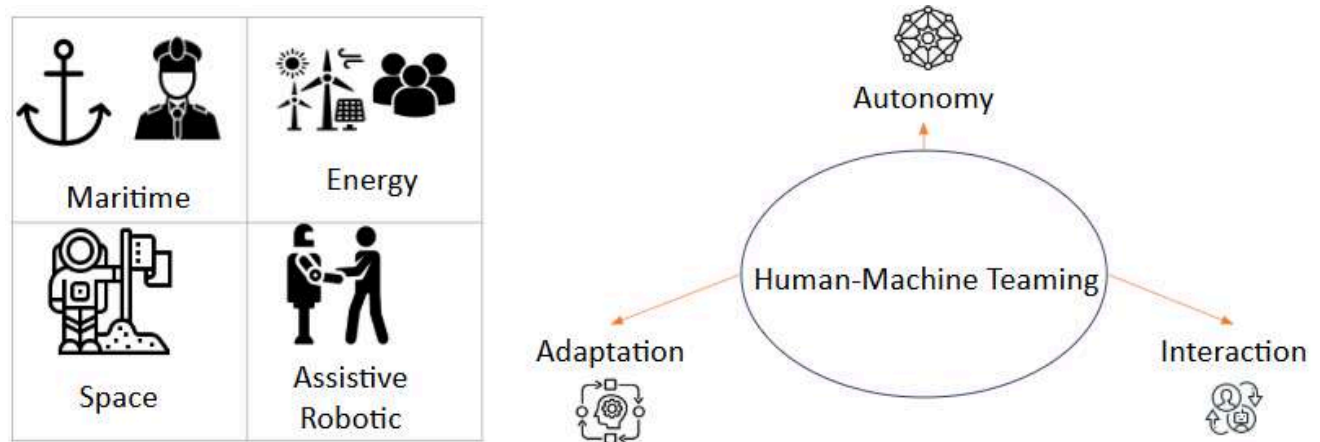


Figure 6: Application Domains for CROSSING 2.0 Research.

1. Maritime

Keywords: Collaborative mission, Multi-Domain Operations, Decision Support Systems

2. Energy

Keywords: Energy Mix, Grid Stability, Smart Grids, Autonomous Operations and Maintenance (O&M), Digital Twins

3. Space

Keywords: Human-Robot Interaction, Situational Awareness, Adaptive Autonomy, Ethical Decision-Making

4. Assistive Robotic

Keywords: Human-Robot Interaction, Cognitive Load Management, Adaptability in Unstructured Environments, Ethical Autonomy, Trust

Maritime**Key Idea:**

In contemporary maritime operations, effective collaboration between manned and unmanned resources is essential, without increasing the workload of operators [Gao & Li, 2020]. The tactical advantages provided by surface or aerial autonomous systems should be extended to underwater operations [Willis & Smith, 2017]. Consequently, underwater drones represent the most promising and demanding environments [Bovio & Caiti, 2014]. In these environments, communications are often limited, sometimes impossible, and occasionally unwanted [Heidemann et al., 2006]. To address these challenges, underwater autonomous vehicles (UAVs) must integrate controlled decision-making systems that enable them to reconfigure their missions and trajectories autonomously in response to evolving tactical situations [Akyildiz et al., 2020]. Achieving this requires the development of on-board artificial intelligence algorithms that can orchestrate a set of functions to ensure the mission is completed automatically, efficiently, and reliably, while adhering to the behavioral rules and limits set by the operator [Baldi & Brunak, 2016]. These systems should be capable of operating on long-endurance, multi-mission drones without the need for constant remote oversight [Benjamin & Curcio, 2012].

Once the challenges of operating underwater are effectively addressed, the path will be paved for tackling multi-domain, with application in collaborative mission scenarios that integrate operations across air, surface, and underwater domains. By achieving interoperability among these environments, forces can enhance situational awareness, optimize resource allocation, and execute more complex and adaptive strategies, ensuring superiority in dynamic and contested battlespaces.

Human operators are responsible for issuing high-level orders, validating planning options, and accessing explanations of the system's decisions at any time [Gueriau et al., 2020]. The system integrates data from various sensors to establish the tactical situation, considering air, surface and underwater information [Thurman & Davis, 2015]. In unforeseen circumstances, such as encountering an enemy ship or an unfavorable environment, the autonomous system should independently react within the given degrees of freedom [Chun & Park, 2021]. AI algorithms enable the system to autonomously adjust its trajectories and behavior as the situation evolves, thereby achieving its objectives [Zhu & Wang, 2020].

The envisioned future naval force comprises a cognitive and adaptive system, composed of collaborative human and artificial agents, alongside both manned and unmanned vehicles. These systems will be equipped with autonomy levels tailored to specific operational contexts, ensuring alignment with doctrines applicable to both land and sea operations [Scharre, 2018]. By combining the strengths of human cognition with advanced artificial agents, these systems will be better positioned to meet the growing demands of modern warfare.

Challenge 1: Tackle limitation of human cognitive capacities

One of the key challenges in high-intensity mission environments is the inherent limitation of human cognitive capacities, which are increasingly strained by the growing complexity of operations and the volume of information to process [Endsley, 1995]. To overcome this, it is essential to design a holistic system that integrates human operators with autonomous agents, ensuring that operational superiority is maintained while adhering to Naval Defense doctrines [Chen & Barnes, 2014]. Ideally human and machine will have a shared mental model of the current situation, or as close as possible, more importantly, and understand how/where their model varies from others involved.

Challenge 2: Collaborative mission rethinks maritime operations as a holistic system

Collaborative mission rethinks operations in terms of effects, moving away from the traditional system-based logic. In infovalorized collaborative mission, all systems are integrated, and the responding element is no longer necessarily the one under attack but rather the one capable of delivering the best effects in the fastest possible tempo. This new paradigm requires:

- Awareness of friendly forces: The goal for the future is to achieve real-time "blueforce tracking", knowing the exact position of all units, monitoring their movements, and coordinating effects.
- Faster operational tempo: Knowing what is happening and who can act, modeling the adversary's maneuvers, and anticipating the effects become crucial.

Challenge 3: Collaborative mission as a large-scale mission including multi operators

In the high-level domain, the aim is to develop decision support systems applied to collaborative naval missions, including multiple operators and autonomous systems from multi-domain (underwater, surface and air) in a large-scale mission. The goal is to use such a system to recommend optimal decisions, including multi operators. Finally, the system will be tested in naval scenarios to ensure it improves decision-making efficiency, supports collaboration between units, and enhances mission success in high-pressure, multi-domain mission operations.

Outcome:

The expected outcome for this use case is a comprehensive framework that allows for human-AI collaboration in maritime operations. By integrating autonomous drones, ships, and human operators, CROSSING 2.0 will provide a holistic solution for maritime operations in real-time. From a computer science point of view, CROSSING 2.0 will develop an architecture coupling heterogeneous sensors from various platforms to perform collaborative actions in a distributed manner, with various operators in the loop and rethinking military operations in terms of effects. This solution will enhance decision-making and operational efficiency, benefiting both defense and commercial sectors.

Energy

Key Idea

Modern power systems rely on a diverse mix of energy sources, including renewable energy, like wind, and solar, or fossil-based ones, like nuclear, and coal. As the energy landscape continues to evolve towards its decarbonization, achieving seamless integration of renewable sources is essential. One of the major challenges in this transition is managing the increasing variability of renewable energy production [Khalib, 2024]. In regards to intermittent energy production, end-user flexibility is one of the key levers for a more renewable-based energy mix. Having humans in the loop generally complicates the controllability of systems. As such, energy systems are getting a bit more complex day by day, uncovering multidisciplinary scientific challenges, and better considering the uncertainty of human behavior.

Challenge 1: End-User Flexibility

The first challenge addressed in CROSSING lies in balancing energy consumption with more intermittent production [D'Ettorre et al., 2022]. This balance requires an improved flexibility of energy consumption. Improving storage solutions (static or mobile ones) are game changer in the way energy consumption can be managed. Flexibility levers can be created at the level of end-users, but also larger (generally industrial) installations. Estimating and activating flexibility levers demands comprehensive data, much of which is gathered through smart meters [Bhela et al., 2018]. Smart grids leverage this data to provide insights into residual energy consumption, as well as real-time grid state, of non-conventional load management [Chukwu & Mahajan, 2014]. However, optimizing the efficiency of smart grids goes beyond simply analyzing available data. It requires strong consumer engagement, meaning the system must not only extract insights from available data but also use that information to develop tangible flexibility mechanisms (incentives) engaging end-users [Cherrelle et al., 2016]. The gathering of rich data on consumer behavior should foster trust in the system, encouraging consumer participation and increasing their flexibility in adapting consumption patterns [Martek et al., 2019]. Developed socio technical solutions should relate technical capabilities with end-user engagement, to provide system operators operational flexibility levers. Currently, however, some limited models effectively capture human behavior in response to flexibility requests, which are crucial for benefiting the grid [Upham et al., 2015]. Human-machine teaming could mitigate the complexity and uncertainty of human-based energy systems.

Challenge 2: Autonomous Operations and Maintenance in Isolated Energy Farms using Human Guidance for Training

The second challenge concerns the sustainability and efficiency of renewable energy farms, particularly in isolated or remote environments. Renewable energy installations, such as offshore wind farms or solar power plants in remote locations, often face harsh conditions, making regular human maintenance difficult and costly. Additionally, these installations must operate autonomously for extended periods, often without immediate access to human oversight or intervention. One promising solution to these issues is the implementation of

autonomous systems for operations and maintenance (O&M) [Piagi & Lasseter, 2006]. Autonomous O&M systems can significantly reduce the operational costs and safety risks associated with human-led maintenance activities in these environments. Autonomous drones equipped with advanced AI algorithms can conduct inspections, perform repairs, and ensure that the systems continue functioning optimally [Bovio et al., 2014]. By utilizing sensors, AI algorithms, and predictive maintenance strategies, these autonomous systems can detect issues before they become critical, minimizing downtime and reducing the need for human intervention. Moreover, AI-driven autonomous systems can adjust their operations in response to real-time environmental data, ensuring that the renewable energy farm operates efficiently under varying conditions [Turner et al., 2020]. A critical aspect of this challenge is the development of robust digital twins for energy systems [Kapteyn et al., 2021]. By simulating environmental conditions and operational scenarios, digital twins can help optimize both the design and maintenance of energy farms, notably in cases where all models (or data) are not available, ensuring they operate effectively even in isolated environments [Dresdner et al., 2018]. Autonomous O&M systems combined with digital twins offer the potential to revolutionize how energy farms are managed, reducing operational costs, improving safety, and ensuring sustainability in challenging environments. Autonomous O&M systems could be trained using human guidance to speed up the process or to set exceptional conditions, such as storms, during training or evaluation.

Challenge 3: Distributed intelligence for the control of complex energy systems

This research axis aims to develop innovative frameworks enabling efficient, resilient, and sustainable operation of energy systems under the increasing complexity of decentralized energy integration, and electrification. As energy systems evolve to accommodate high shares of renewable energy, distributed energy resources (DERs), and electrified sectors, traditional centralized control architectures are indeed becoming inadequate. Distributed intelligence, facilitated for instance by multi-agent systems (MAS), offers a scalable and adaptive solution to manage the intricate dynamics of these systems while ensuring optimal performance, stability, and security. The main questions concern the effective distribution of intelligence to enhance decision-making in real-time, resource-constrained energy environments. In this context, the selection and implementation of algorithms and protocols for the coordination of agents in such systems constitute scientific challenges facing cyber-physical uncertainties, such as demand variability, renewable intermittency, but also cyber threats [Howell, 2017]. Approaches will address the development of distributed optimization models, like energy scheduling including market mechanisms and grid constraints for fair resource allocation, but also the design of multi-agent coordination frameworks, equipping them with machine learning capabilities to adapt to changing system conditions. Digital twins will be helpful to test and validate the proposed models under realistic scenarios, including high penetration of renewables, DER proliferation, and grid contingencies.

Outcome:

This research is expected to deliver a comprehensive framework for the integration and management of renewable energy systems. By addressing end-user flexibility, autonomous operations, and distributed intelligence, it aims to enhance the efficiency, sustainability, and resilience of energy systems in increasingly complex environments. The framework will leverage smart grids, AI-driven predictive maintenance, and multi-agent systems to optimize energy consumption, ensure reliable operation of isolated energy farms, and enable adaptive, real-time decision-making. From a technological perspective, the research will pioneer advancements in digital twins, predictive modeling, and distributed optimization, setting new standards for the management and control of decentralized, renewable-driven energy systems.

Space

Key Idea:

Deep space habitation for long periods of time will generate exciting challenges for human-robot interaction and human-machine teaming more broadly. Human-robot interactions on Earth can take place in the form of a robotic vehicle that is remotely operated by a human (teleoperation), or supervised by a human (supervisory control). In these scenarios, the human plays a critical role in the success of the robot's activity and they work cooperatively in complex environments. In space, there will be new challenges for both near real time and delayed supervisory control for complex tasks in a harsh environment, in terms of maintaining situational awareness and trust. Human-robot interaction will be a key part of a higher proportion of activities off Earth, due to the harsh environment. We therefore also consider challenges associated with human-robot interaction over extended periods of time for remote teams.

CROSSING 2.0 will work in association with the Andy Thomas Centre for Space Resources at the University of Adelaide. This use case already involves a PhD topic, cofounded by the EU SEED program administered by IMT-Atlantique and the University of Adelaide. CROSSING 2.0 will also participate in grant applications to extend the facilities at Roseworthy which will support testing of human-robot interaction in the field, as well as new forms of AR and VR while walking freely. Researchers are working on an outline of a program of research to develop new types of AI algorithms that are able to learn about a context with limited prior training, and can support human-robot interaction. This approach will include developing new ways for the human and robot to maintain common understanding of a task as a form of Space Domain Awareness (SDA). Another important aspect is that a part of the work is the control or the interaction with a group of robots operating in harsh environments and limited or delayed communications with human operators.

Challenge 1: Communication and Trust in Human-Robot Interaction

Maintaining situational awareness and trust over long distances and time delays is critical. Robots in space must anticipate human intentions, respond to indirect cues, and operate with minimal supervision while retaining human operators' confidence. This requires developing robust communication protocols and trust-building mechanisms that support transparent, intuitive interactions in unpredictable environments.

Challenge 2: Managing Cognitive Load on Human Operators

Extended space missions may risk overwhelming human operators, particularly if robotic systems demand high levels of oversight. By designing intuitive interfaces that balance autonomy with transparency, this research aims to reduce cognitive load, allowing operators to focus on high-priority tasks. Adaptive autonomy levels enable robots to dynamically engage humans only when needed, maintaining efficient collaboration without overburdening operators.

Challenge 3: Adaptability in Unstructured Environments

Space presents an unstructured, high-risk environment that robots must navigate autonomously, encountering unexpected obstacles and adapting to evolving task priorities. Developing

resilient, adaptable systems that can handle such unpredictability is essential for reducing reliance on direct human intervention and ensuring mission success.

Challenge 4: Ethical Considerations in Human-Robot Teaming

As robots gain autonomy, ethical implications regarding safety, privacy, and decision accountability become central. Robots must align their actions with ethical guidelines, particularly in high-stakes scenarios. The research program will explore frameworks for accountability and transparency in robot decision-making, building trust and ensuring responsible, safe interactions over extended periods.

Outcome:

The expected outcome of this research is a comprehensive framework for human-robot collaboration in deep space missions. By integrating autonomous systems capable of dynamic task allocation, adaptive mission planning, and responsive communication, CROSSING 2.0 aims to enhance decision-making and operational efficiency in space. This architecture will foster trust and shared understanding in human-robot teams, ensuring autonomous agents provide meaningful support across complex tasks in unpredictable conditions. From a computer science perspective, CROSSING 2.0 will develop an integrated solution leveraging multi-agent systems, predictive modeling, and adaptive control, facilitating robust human-robot teaming in space exploration and establishing standards for autonomous collaboration beyond Earth.

Assistive Robotic

Key Idea:

Assistive robotics research in CROSSING focuses on enabling robots to engage with humans in real-world environments. This line of research emphasizes building intuitive interfaces and control mechanisms for robots that allow for seamless collaboration with humans. The goal is to achieve advanced levels of human-robot interaction, where robots not only respond to human commands but also anticipate needs and provide proactive support.

This research is centered on creating socially intelligent robots that leverage verbal, non-verbal, and para-verbal communication to assist individuals. These robots aim to improve users' quality of life and work by offering support for various tasks and continuous monitoring. Key focus areas include user modeling, adaptive learning, and behavior control, all tailored to meet the unique needs of individuals. Scientific contributions will benefit from the RAICAM (<https://raicam.eu/#home>) and SWEET (<https://www.sweet.unina.it/>) ongoing European projects, including CROSSING 2.0 partner in their consortium.

Challenge 1: Communication and Trust in Human-Robot Interaction

One of the key challenges in human-robot teaming is ensuring smooth communication and trust between humans and robots [Simon et al., 2023]. Robots must interpret human intentions, adapt to dynamic environments, and predict the needs of their human partners, even when direct communication is limited or absent. Robots need to understand and respond to indirect cues such as gestures or body language. Establishing trust is vital, as effective collaboration depends on the human operator's confidence in the robot's capabilities.

Challenge 2: Managing Cognitive Load on Human Operators

As robots take on more sophisticated roles, one of the major challenges is managing the cognitive load placed on human operators. When robotic systems require constant supervision or when their actions are opaque, human operators can become overwhelmed. This could lead to over-reliance on the robot, where humans may relinquish too much control, or under-reliance, where humans avoid using the robot's full capabilities. Designing user interfaces that balance autonomy and transparency is essential to reducing cognitive load and improving collaboration efficiency [Dragan et al., 2013; Garcia et al., 2024].

Challenge 3: Adaptability in Unstructured Environments

Robots often need to operate in highly unpredictable and unstructured environments. In situations such as disaster response, defense operations, or warehouse logistics, robots must adapt quickly to unforeseen challenges such as obstacles or changing task priorities. The ability to operate autonomously in these environments, without the need for constant human intervention, is critical to the success of human-robot teams. Developing robust and resilient

systems that can handle such unpredictability is a central challenge in assistive robotics [Lyons, 2013].

Challenge 4: Ethical Considerations in Human-Robot Teaming

As robots gain more autonomy, the ethical implications of their actions become increasingly significant [Lin et al., 2011], particularly in sensitive areas like healthcare and defense. Ensuring that robots make ethically sound decisions while respecting human privacy, autonomy, and safety is a crucial challenge. Furthermore, accountability for the robot's actions must be clearly defined, and safeguards must be put in place to ensure that robots act responsibly and transparently [Goodrich & Schultz, 2007]. These considerations are essential for building trust and ensuring the long-term viability of human-robot collaboration.

Outcome:

The outcome of this research will be a set of assistive robotic systems capable of adapting to human behavior in real-time, enhancing efficiency in shared human-robot tasks. These systems will be tested in real-world environments such as hospitals, warehouses, and defense settings, where human-robot collaboration can lead to increased operational efficiency and safety. The research will also contribute to new models of human-robot teaming, focusing on reducing cognitive load, increasing transparency, and addressing ethical challenges in autonomous robotic systems.

3. Impact, Human Resources and Facilities

Australia and France Benefits

The **IRL CROSSING** lab offers numerous benefits to both Australia and France by driving forward cutting-edge research in human-AI collaboration and autonomous systems:

- **Economic Impact:** By fostering collaborations with industries such as defense, healthcare, and energy, the lab promotes innovation and competitiveness. Naval Group's involvement enhances Australia's defense capabilities, while contributing to France's technological leadership in autonomous systems.
- **Scientific Collaboration:** CROSSING bridges expertise from leading French and Australian universities, promoting cross-cultural and interdisciplinary research. This partnership strengthens global ties in AI, robotics, and human-machine interaction, advancing scientific knowledge and technology in both countries.
- **Workforce Development:** The lab provides unique training opportunities for PhD students and researchers, fostering a new generation of experts in AI and robotics. These experts will play pivotal roles in the development of critical technologies, benefiting both nations' labor markets.
- **Strategic Applications:** With a focus on industries like maritime surveillance and energy management, CROSSING's research contributes to national security and sustainable energy solutions, aligning with both countries' strategic priorities.

These collaborations support high-level innovation while reinforcing the global positioning of France and Australia in AI and autonomous technologies.

Long-term Mobilities

The following table and figure provide a tentative overview of the mobility forecast for CROSSING 2.0, valid as of 09/01/2025.

Name	Institution	Delegation Dates	Period	On site before 2026
Paul Egre	CNRS	2026	9 months	✓
Vincent Debusschere	UGA	2026	9 months	✓
Thibault Gateau	ISAE-Supaero	2026	6 months	✓
Nicolas Drougard	ISAE-Supaero	2026	6 months	
Emmanuel Chemla	CNRS	2026-2027	1.5 years	✓
Cedric Buche (director)	IMTA	2026-2030	5 years	✓
Scientist X	Naval Group	2026-2030	5 years	✓
Scientist Y	Naval Group	2026-2030	5 years	✓
Tarek Hamel	Université Côte d'Azur	2027-2029	2 years	
Christophe Viel	CNRS	2028-2029	1 to 2 years	
Lucia Bergantin	IMTA	2028-2029	1 to 2 years	
Benoit Clement	ENSTA	2029-2031	1 to 2 years	
Gilles Le Chenadec	ENSTA	2029-3030	6 months to 1 year	

Table 4: Long-term Mobility Plan for CROSSING Researchers.

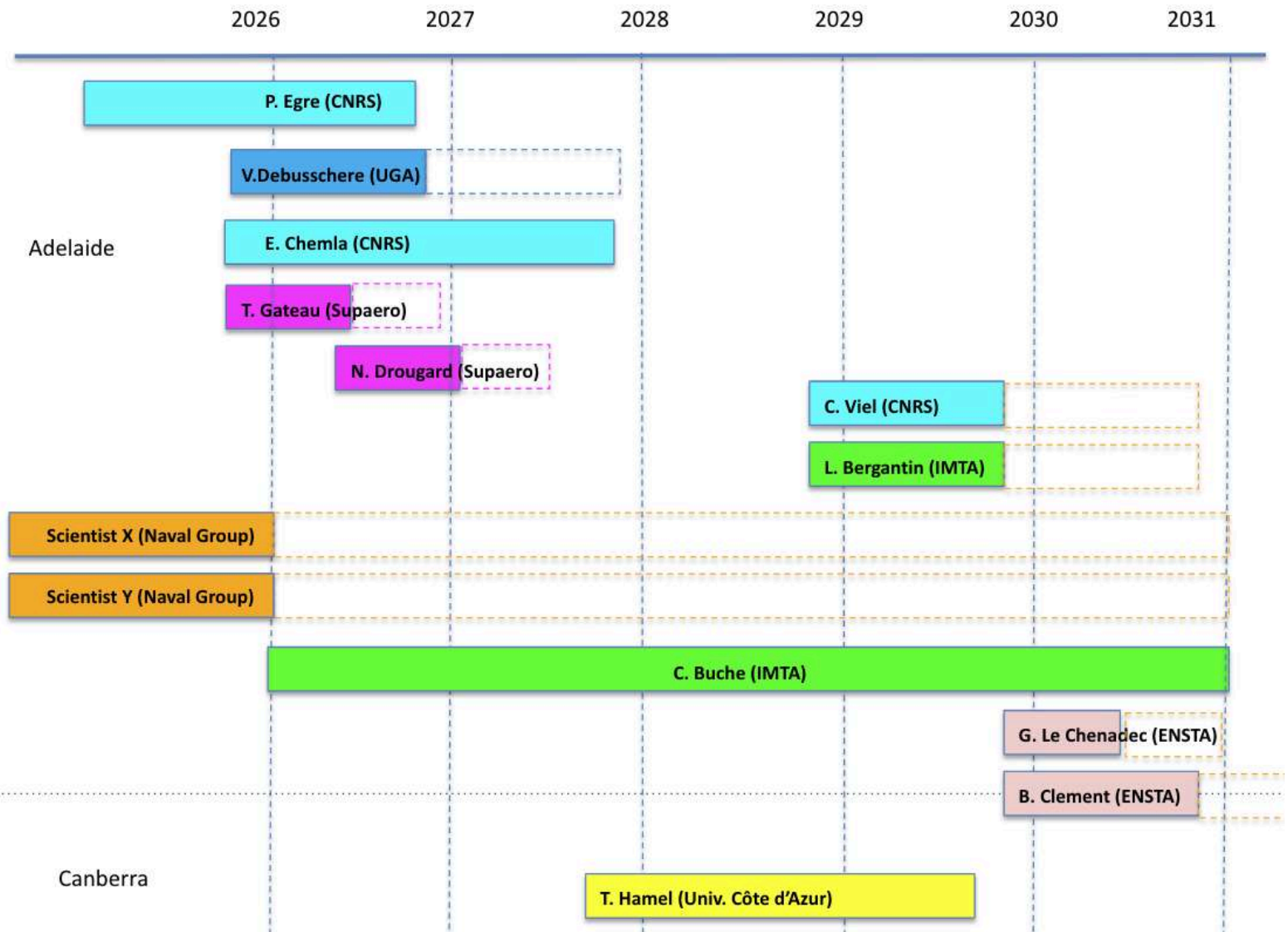


Figure 7: Mobility Forecast for CROSSING 2.0.

Experimental Facilities in Australia

Beside central fundamental aspects, CROSSING's research is also devoted to addressing the dimension of valorization and therefore demonstrates transfer capacities to major industrial and applicative partners. Even if CROSSING members' direct activities may obviously not cover the whole spectrum of technological transfer, we plan to facilitate this kind of initiative while relying and demonstrating research results on platforms.

Different platforms, digital and cyber physical ones, may be shared within the laboratory to constitute as many sandboxes allowing to develop, deploy and demonstrate news concepts issued from CROSSING's research. We can already mention the following platforms.

Name	Partners	Location	CROSSING Use case
Maritime Lab	Flinders University	Australia	Maritime
LOTUSim	Naval Group	Australia / France	Maritime
Andy Thomas Roseworthy	Adelaide University	Australia	Space
IVE	Adelaide University	Australia	Maritime / Space
BBB	Adelaide University	Australia	Maritime / Space
Active Vision Lab	Adelaide University	Australia	Maritime / Space
School of Cybernetics	ANU	Australia	Maritime / Space / Assistive Robotics

Table 5: Experimental Facilities Supporting CROSSING Research.

In addition, various assistive robots are available in both countries.

Maritime Laboratory - Flinders University

The Maritime Laboratory, located at the Flinders campus, provides a wide range of experimental support. This includes underwater vehicles (AUV, ROV), surface vessels (WAM-V, BlueBot), and aerial drones, allowing the evaluation of multi-agent, and potentially multi-domain missions. The laboratory is also equipped with a water tank of 6 by 6 meters with a maximum depth of 3 meters. An external camera-based localization system is able to track the vehicle when evolving underwater, allowing the evaluation of path planning, trajectory following, or station-keeping tasks. In addition, scanning sensors are also available (e.g. embedded cameras or side-scan sonars) allowing to simulate scenarios where robotic systems have to collaborate with human operators during their mission.

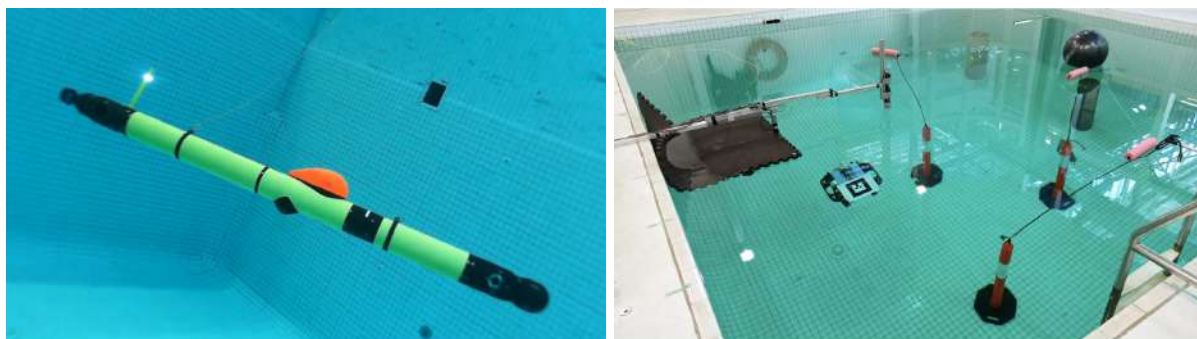


Figure 8: Maritime Laboratory at Flinders University.

LOTUSim - Naval Group

CROSSING's maritime research is unified through the LOTUSim platform, a real-time multi physics and multi-agent simulation system developed by Naval Group. This platform provides a realistic environment for testing human-drone teaming across air, surface, and underwater domains. Operators will interact with these simulations through immersive interfaces like AR/VR. Simulation platforms like LOTUSim enable comprehensive multi-agent system training and provide realistic physics models that enhance AI learning. Real-time simulations allow for stress-testing AI systems under real-world conditions, such as environmental fluctuations and complex, dynamic scenarios.

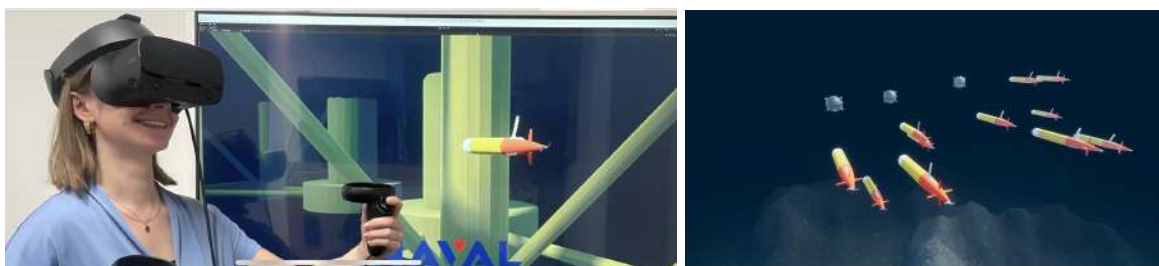


Figure 9: LOTUSim Simulation Platform for Maritime Research.

Andy Thomas Roseworthy facilities and space applications - Adelaide University

The Andy Thomas Centre for Space Resources (ATCSR) brings together collective exploration, mining, manufacturing and engineering research strengths to address the challenges faced by long term planetary exploration, while ensuring the near-term application here on Earth. The new facility is a 4,000m² open-field robotic testbed, operations centre and workshop facility, designed to provide a complete solution for scaled to full-sized rover testing and development. Located 50 mins north of Adelaide at the University's Roseworthy Campus, the new facility offers testing and development scenarios across a range of simulated off-Earth environments and settings, including investigations in human-robot teaming and robotic perception.



Figure 10: Andy Thomas Centre Facilities for Space Research.

Australian Research Centre for Interactive and Virtual Environments - Adelaide University

The Australian Research Centre for Interactive and Virtual Environment (IVE), based at the Mawson Lakes campus, is a multidisciplinary research centre of over 140+ researchers from creative, architecture, design, culture and anthropology, digital preservation, construction, project management, industrial design, psychology, neuroscience, fatigue, electronics and prototyping, wearables and sensor devices (including EEG), computer science, and virtual and augmented reality. The centre focuses on the relationship between humans and technology, working across Defence, health, education and training, manufacturing, digital twins, design, and agriculture. With access to the latest virtual and augmented realities technologies, IVE works at the cutting edge of human-machine interaction. The centre has a number of unique facilities, including large scale motion capture facilities, 3D volumetric capture spaces, augmented reality sandbox, spherical cave, VR treadmills, as well as a purpose built, large-scale projected augmented reality space. The projected AR facility is 10x7m.

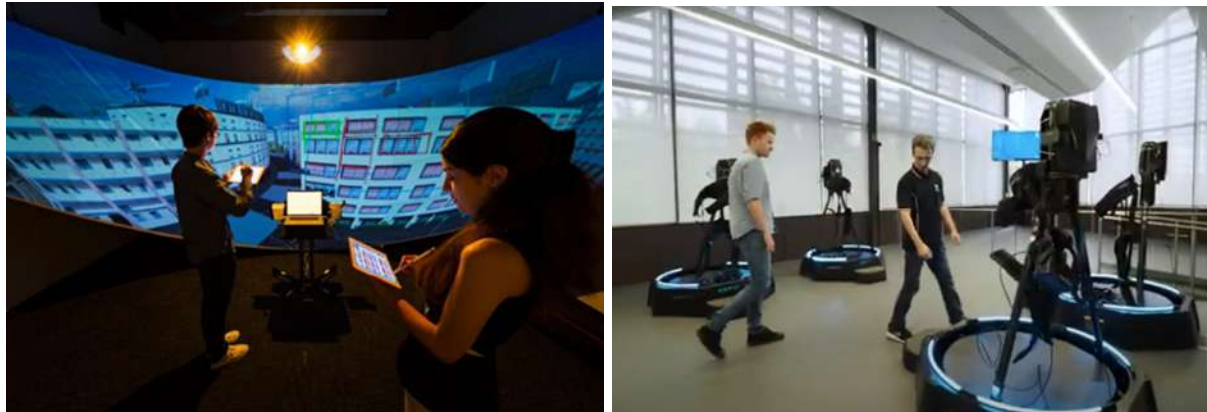


Figure 11: Australian Research Centre for Interactive and Virtual Environments.

Active Vision Lab - Adelaide University

The Active Vision Lab studies how people use visual information to interact with their environment, and how attention is deployed as people interact with their environment. Missions are 1/ work with patients with clinical field loss as well as normally sighted groups to understand how disease and healthy ageing impact active vision and sensorimotor control and 2/ investigate how active vision (including deployment of attention) is impacted during human-autonomy teaming and work on how to build more effective human machine interfaces.



Figure 12: Active Vision Lab Research Overview.

Behaviour-Brain-Body Research Centre - Adelaide University

The Behaviour-Brain-Body Research Centre (BBB) investigates people through the combined lens of behaviour, brain and body. A team of over 30 experts works collaboratively across behavioural change, fatigue and stress management, decision-making and recovery, to predict,

prevent, detect and intervene, all with the aim of optimising human function. At BBB we blend pure research with real world contexts to help our community, government and industry partners tackle their thorniest questions and create solutions that support behavioural resilience and promote healthy functioning. The enterprising way we engage with our partners pivots on our ability to bring together people with different expertise to solve these big challenges.



Figure 13: Behaviour-Brain-Body Research Centre Activities.

Assistive Robotics

CROSSING benefits from various robots, some are available in France and in Australia, facilitating exchanges and mirroring for development and experiments.



Figure 14: Assistive Robots Available for CROSSING 2.0 Experiments.

School of Cybernetics - ANU

The ANU School of Cybernetics hosts a Maker Space for fast prototyping purposes. The space is equipped with:

- Video and audio recording equipment
- 360 digital camera
- VR headsets
- Robot Arm - Kinova Gen3 DoF
- Robot – NAO, 3D printer
- soldering equipment
- wet lab, etc.



Figure 15: ANU Facilities for Experiments.

4. Research Mapping

The following table outlines a strategic research map centered on the three core scientific axes, detailing their alignment with the four use cases and the targeted facilities that serve as test beds for validating and advancing research outcomes.

Axis \ Use Case	Maritime	Energy	Space	Assistive Robotics
Autonomy	Swarm of drones missions (coverage, energy, mapping, maintenance ...)	Smart grid Swarm of drones: operations & maintenance	Autonomous spacecraft operations	Help at home (navigation in an unknown environment, safety, grasping ...)
Interaction	Mission control interface	Efficient management of complex energy networks Operations and Maintenance Monitoring	Teleoperation Supervisory control	Dialogue Engagement (Back channel) Beaming Non-verbal and para-verbal communication Haptics and Touch
Adaptation	Controlled decision-making autonomy (ADC)	Real-time adaptation: - nudge/incentive - based consumer choices - production / storage decisions Chaos Engineering for Resilient Energy Supply Chains	Trust in remote condition Task with delays	Cobot - share tasks
Test bed	Maritime Lab Tank DSA LOTUSim IVE BBB	Experiment'HAAL Active Vision Lab	Andy Thomas Centre Active Vision Lab BBB IVE	Experiment'HAAL IVE

Table 6: Strategic Research Mapping Across Use Cases.

Appendices

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Note: CROSSING members' references are labeled with (*)

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Table 7: Document Contributors for CROSSING 2.0 proposal.

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Table 9: Institution Validators for CROSSING 2.0 proposal.

Members and Associated

Members vs Associated

Members are individuals formally affiliated with the lab, typically working on-site as full-time or part-time researchers directly contributing to the lab's activities.

In contrast, associates are not formally employed or integrated into the lab's structure but maintain a collaborative relationship. Associates may occasionally work on-site during visits. If an associate remains actively involved on-site for a continuous period of at least one year, their status may transition to that of a member.

Aspect	Member	Associate
Commitment	Full-time or part-time involvement on sites	Project-specific or occasional involvement.

Table 10: Members definition.

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Table 11: Members List.

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French

Name	Affiliation	Position	Visit Forecast
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Emmanuel Chemla	CNRS	Senior Research Scientist (DR)	2025-2027 (2 years)
Benjamin Spector	CNRS	Senior Research Scientist (DR)	Nov 2025 (one month)
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Table 12: Associated List (French).

Australian

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Table 13: Associated List (Australian).

Experimental Facilities in France

The following section of the document focuses on experimental facilities in France, complementing those in Australia, and is particularly valuable in the context of joint supervision agreements and co-located projects.

Name	Partners	Location	CROSSING Use case
Tank	ENSTA	France	Maritime
DSA	Naval Group	France	Maritime
Experiment'HAAL	IMTA	France	Assistive robotic
CEXI-HA	IMTA	France	Maritime / Space
SENI	Naval Group / ENSTA	France	Maritime

Table 14: Experimental Facilities in France.

Tank for testing robots and swarms of autonomous drones - ENSTA

This is a new step up in terms of extending testing capabilities for marine robotics for exploratory purposes (on the surface, underwater and aerial). Research projects are experiencing strong growth in these areas. These marine robotics facilities will be unique in France and sized to accommodate the different types of marine drones and exploratory methods developed at ENSTA. This new site will provide optimal conditions for validating the detection, localisation and coordination algorithms of autonomous robots. Measuring 20 meters long, 12 meters wide and 6 meters deep, this test tank will be built into a specific structure that lets through GPS waves (timber framework and textile membrane). Thanks to the addition of an overhead flight area, it will be possible for the marine robots in the tank to communicate with flying drones. The site is also expected to have a bridge crane, for deploying prototypes, a workshop area overlooking the tank and a storage area. These facilities will be managed by the Resource Center of ENSTA. Work on this site began in 2024, with a demolition phase of the old building and laying down of deep foundations. The structural works will commence in mid-September 2024. Delivery is scheduled for the end of 2025.



Figure 16: Naval Group's DSA Test Facility.

DSA - Naval Group

Sirehna has privileged access to the sea thanks to the St-Mandrier site in the south of France to carry out tests on its prototypes and the Naval Group prototypes in real-life conditions. The secure test site is composed of a 500m² shed with partnerships and local infrastructures to facilitate testing. This marine base enables the development of activities: Naval Group has recently completed a sea demonstration of its demonstrator XL UUV and Sirehna is qualifying his new USV, the SEAQUEST.



Figure 17: Experiment'HAAL Platform for Assisted Living.

Experiment'HAAL - IMTA

The Experiment'HAAL (Human Ambient Assisted Living) platform was certified by the Forum des Living Labs en Santé et Autonomie in December 2013. Intended to develop and then host experimental assistance devices for use tests, this platform allows in-situ testing of services developed in research projects before their deployment in real-life situations. Experiment'HAAL is made up of 4 rooms:

- Apartment: A room representative of a residential habitat including kitchen, bathroom, living room, dining room, hosts furniture, sensors, actuators.
- Engine room : A 15 m² room, the technical room, which hosts servers, workstations.

- Control room: This 24 m² room is dedicated to user tests. The one-way mirror is particularly well suited to user tests in conjunction with ergonomists and sociologists.
- Robotics room: This 34 m² room is used for work on several types of robots.



Figure 18: CEXI-HA Research Facility.

CEXI-HA - IMTA

The CEXI-HA facility provides a comprehensive environment for advanced research on augmented humans. The facility includes:

- Two OptiTrack tracking spaces with 16 cameras each
- VR/AR/XR headsets, including HTC Vive and Vive Cosmos, Meta Quest 2 and 3, Varjo XR3, HoloLens 2, and Magic Leap 2.
- Large touchscreen displays (two fixed and two mobile, with adjustable inclination).
- Projection capabilities for immersive displays.
- Eye-tracking technologies for behavioral studies.
- Brain-Computer Interface (BCI) equipment,
- Physiological sensors

SENI - Naval Group / ENSTA

Naval Group and ENSTA have, over the years, identified significant scientific and technical challenges associated with designing and developing intelligent naval onboard systems. To address these challenges, they established a key initiative: the SENI common laboratory (Intelligent Naval Embedded Systems), launched in April 2018 under a collaboration framework agreement.

ANU - Integration Proposal



Australian
National
University

Proposal to join IRL Crossing

French-Australian Laboratory for Humans &
Autonomous Agents Teaming

2026-2030

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Context

International Research Laboratory, IRL.

An IRL is the highest cooperation mechanism deployed by the CNRS (French National Centre for Scientific Research)¹. This is a Joint physical lab that brings together researchers, PhD students, postdocs, engineers, and technicians from CNRS and from partner institutions.

IRL Crossing

IRL Crossing² is a multidisciplinary Australian French Lab for Human-Machine Interactions and Teaming. The lab had a first mandate of 5 years, which comes to an end in December 2025. Partner institutions of the first iteration of the joint lab are: the University of Adelaide, the University of South Australia, Flinders University, CNRS, IMT Atlantique (French engineering school), and Naval Group as an industry partner³.

In 2024, IRL Crossing comprises about 18 members, 8 associate members and 20 PhD students including: 12 France-Australia Joint PhD students, 5 PhD students funded by Naval Group, and 3 PhD students funded under Aufrande (European Cofund). Also involved in the project are 10 master's students and interns. Co-directors are Jean-Philippe Diguët (CNRS) and Anna Ma-Wyatt (University of Adelaide).

IRL Crossing is now in the process of renewing its mandate for 5 years – 2026 to 2031. There is a willingness from current members of the lab and research groups from the ANU School of Cybernetics and the ANU School of Engineering to join forces for this new iteration.

Opportunities

Deepening ANU's partnership with CNRS on AI and robotics

ANU is currently collaborating with CNRS through the project IRP-ARS (International Research Project – Advanced Autonomy for Robotic Systems)⁴, which is led by Professor Robert Mahony from the ANU School of Engineering and Professor Tarek Hamel from University of Côte d'Azur. The project is a partnership between the University of Côte d'Azur, the University of Lorraine, the ANU and the University of Melbourne.

This project hosts 8 PhDs under joint supervision. As the project is coming to an end in 2024, the idea would be to have this project merge with IRL Crossing as well.

Additionally, the School of Cybernetics (formerly 3Ai) has a smaller joint ANU-CNRS collaboratively funded initiative on algorithmic decision theory linked to policy design with CNRS-LAMSADE to support PhD student exchange, due to end in December 2024. A number of the themes, and inter-disciplinary sets of expertise developed for them, are relevant to the next iteration of IRL-CROSSING.

Access French and European funding

All members of the IRL, as full members of a CNRS Joint Unit, are eligible to funding from the French National Research Agency (ANR) and from the European Union.

¹ The CNRS has a representation office at the University of Melbourne:

<https://melbourne.office.cnrs.fr/about-the-cnrs/>

² <https://crossing.cnrs.fr/>

³ Naval Group is a European leader in naval defence design, development and construction and dismantlement.

⁴ <https://melbourne.office.cnrs.fr/project/irp-ars/>

Internationalise the College

Joining IRL Crossing would allow ANU students and faculty members to benefit from the following opportunities:

- Internships for undergraduate and postgraduate students, for a minimum of 6 weeks of stipend.
- Grants for PhD students and postdoctoral fellows to support costs associated with their research, which could include conference travel, travel between France and Australia for scientific collaboration, small equipment and publication costs, or other associated costs.
- Grants to support visits from academics for short and long stays.
- Small funds for new projects of IRL Crossing members, to support data collection or the establishment of new collaborations.

Scientific project

With this joint lab, members will contribute to different streams of research, focusing on both social science and engineering approaches to robotics and human-machine teaming. The streams are listed below in no particular order.

Stream One: Advancing Autonomy for Uncrewed Robotic Systems

This stream of research aims to advance the fundamental science and technology underlying individual and networked uncrewed systems.

The technological capacity to build physical machines capable of carrying out the required tasks exists. However, a significant roadblock to deployment lies in the robustness and reliability of perception algorithms, that observe and interpret the infinitely complex world in real-time, and of control algorithms, that safely and efficiently guide a robotic system through this complexity to achieve its goals. Moreover, these machines often have complex and poorly known dynamic models. The sensor systems used can be very noisy and poorly characterised. The applications considered may require them to operate closer to the vehicle performance limitations than for crewed vehicles. They might often evolve in close proximity to an unknown (or partially known) and dynamically changing physical environment.

The goal is to go beyond the state of the art in systems and control theory applied to Uncrewed Robotic Systems. A number of theoretical open problems with high impact in practical applications will be addressed to enable Uncrewed Robotic Systems to operate more robustly and more reliably in complex and cluttered dynamic environments encountered in real-world applications.

The progress made during the previous IRP-ARS period provides a strong foundation for the next stage of research. In particular, the team is focused on leveraging the non-linear and symmetry-based insight developed to provide solutions to the still open questions of combining high-performance control with real-time and reliable state estimation and situation awareness.

Stream Two: Applying Neurodesign to the Creation of Sophisticated Interactive Systems

This stream of research lies at the intersection of human-technology interaction, human-machine teaming, as well as both human and machine prediction and decision-making. Here, we will explore mechanisms of human perception and learning and how these cognitive processes can inform the development of sophisticated interactive systems. Our research team refers to this approach as ‘neurodesign,’ a methodology that our team has previously applied to the

creation of mixed digital and physical experiences that incorporate technologies such as virtual reality, augmented reality, and artificial intelligence. Additionally, this stream of research will be exploring the implications of these advancements within speculative and applied futures, aiming to contribute to the ethical and purposeful integration of technology in human environments.

Stream Three: Agile governance and policy design for Autonomous Agent/Human Interaction

Building on the research of the first IRP-ARS period, this stream aims to extend the new paradigms and solutions to focus on unresolved tensions of governing and policy environments within which human-machine teaming is increasingly found, including defence. Here we will explore challenges and opportunities of algorithmic and human governance in fast-changing environments (due to technology upgrades, new research, environmental and social conditions, as well as new learned behaviours of both humans and autonomous agents), including the development of analytics for measuring risks and opportunities of deployment, and skills for leading change and responsible management of hybrid teams in complex settings. This stream of work will build on the state of the art in algorithmic decision theory and policy design, agile governance and cybernetics/complexity leadership, aiming to contribute research to inform the governing structures, rules, policies and operational guidance systems (including human and algorithmic training) required for successful navigation of complex environments.

IRL Crossing current focus areas

During its first and current iteration, IRL Crossing focused on four priorities areas for research:

- 1. New Models to Understand and Anticipate Human Behaviour**
 - 1.1. Capture of Human Physiological, Cognitive Performance and Emotional State
 - 1.2. Using Human Data to Augment Individual and Team Decision Making, and Team Management
 - 1.3. Human Activity, Capability and Performance Understanding
- 2. New Algorithms for Energy-Efficient and Human Based AI**
 - 2.1. Cross Learning Between Humans and Machines
 - 2.2. Self-Adaptation for Cooperation
 - 2.3. Distributed AI on Resource Constrained Devices for Real-Time Performance
- 3. New Paradigms for Autonomous Agents/Human Interaction and Understanding**
 - 3.1. Intelligent and Interactive Autonomous Agents
 - 3.2. Theoretical Models of Interaction and Human-Machine Dialogue
 - 3.3. New Interaction Frameworks and Technologies to Facilitate Cooperation within a Hybrid System
- 4. New Solutions for the Management of Hybrid Teams**
 - 4.1. New Models of Human-Intelligent Machine Cooperation and Interaction
 - 4.2. Adaptation and Regulation to Reach Cognitive Equilibrium Between Humans/Autonomous Agents

New research priorities will be developed for the second iteration of IRL Crossing. *ANU will be able to shape them so that they align with our own research strategy.*

Resources

To join IRL CROSSING as a stakeholder (“tutelle”), ANU will have to contribute resources. This will need to be negotiated. Currently, each IRL CROSSING university member contributes:

- The equivalent of 1 full-time research staff member: 0.5 for a Professor, and the rest can be split into 0.1. Partners usually list between 3 to 6 permanent members to the lab.
- 2 funded PhD students, to start on the 1st and 2nd year of the contract.
- A contribution of 50,000 AUD per year (can be in kind, like equipment of office space).
- Professional staff if needed (0.5 or less).

Partner resources are available to the whole lab and to all members, regardless of their affiliation.

The ANU, College of Engineering, Computing and Cybernetics, offers to:

- Affect at least 6 permanent members to the lab under part-time contribution arrangements.
- Supervise and fund 2 PhD students, under the current internal funding rules of the university.
- Provide office space as well as lab access, and where appropriate research support (e.g. lab technician/Research assistant support), amounting to 50,000 AUD per year.

Calendar

November 5: Antoine PETIT, CEO of CNRS and Genevieve Bell, VC of ANU meet to sign a partnership agreement to open-up a joint lab (IRL) focusing on mathematics. This is an opportunity for both ANU and CNRS to discuss the potential alignment of interests outlined here, and to confirm the in-principle support of ANU and CNRS to collaborate on IRL Crossing.

November: Presentation of ANU’s proposal to join IRL Crossing as a stakeholder member, to IRL Crossing steering committee.

2025: Shape the scientific project and negotiate the agreement terms.

June 2025: Sign the agreement.

January 2026: Launch of the new IRL Crossing lab.

Interested members

Prof Katherine Daniell, Interim Director of the School of Cybernetics

Trained in engineering, arts and public policy, Katherine’s work focusses on collaborative and inclusive approaches to policy, action and education for sustainable development. Katherine delights in the clash and novel synthesis of disciplinary cultures and uses cybernetics to convene challenging and productive conversations.

As a transdisciplinary academic, she has worked in Australia and globally on areas including international science and technology cooperation, governance, risk, leadership, AI systems, river basin management, politics and cultures of innovation, and climate change adaptation.

She has produced over 100 academic publications including 4 books and a diverse range of book chapters, papers, reports, and edited collections, and has graduated over a dozen PhD students. Katherine was the Inaugural Convenor of the Master of Applied Cybernetics program from 2019-22.

Katherine is also currently a John Monash Scholar, Member of the National Committee on Water Engineering, Editor of the Australasian Journal of Water Resources, member of the Rivers Committee of the Initiatives of the Future of Great Rivers, President of the French-Australian Association for Research and Innovation (AFRAN) Inc., and a Chevalier (Knight) in the French Ordre National du Mérite.

Prof Robert Mahony, Professor and IEEE Fellow, Systems Theory and Robotics Group, School of Engineering

Robert Mahony is a Professor in the Research School of Engineering at the Australian National University. He received his BSc in 1989 (applied mathematics and geology) and his PhD in 1995 (systems engineering) both from the Australian National University.

He is a fellow of the IEEE and was president of the Australian Robotics Association from 2008-2011. His research interests are non-linear systems theory with robotics and computer vision applications. He is known for his work in aerial robotics, geometric observer design, matrix subspace optimisation, and image-based visual servo control.

The evidence his research is considered world-leading is reflected in his citation count (25200), his h-index of 61 (Google Scholar), and his plenary and keynote talks at leading international and national conferences. These include the International Conference on Robotics and Automation (ICRA), IFAC Symposium on Nonlinear Control Systems (NOLCOS), International Symposium on Robotics Research (ISRR), Asian Control Conference (ASCC), Engineering Mathematics and Applications Conference (EMACS), International Conference on Control and the Automation and Information Sciences (ICCAIS).

Professor Alex Zafiroglu, ANU Futures Fellow and Associate Director HDR, School of Cybernetics

Professor Zafiroglu's work focuses on temporality in socio-technical systems informed by systems theory across cybernetics and social sciences.

Professor Zafiroglu's current research, the *Cybernetics of Rapport and Attentiveness*, addresses how we can learn to recognise the temporal rhythms, scale and logics that always and already structure our 'here and now' as we attend to, and are attended by, computational systems.

Professor Zafiroglu established the PhD Program in the School of Cybernetics in 2020, which is characterised by a concurrent approach, a commitment to realism in the skills and expertise students bring to cybernetics, and the access to experienced professors and cybernetic mentors and skills during the course of study.

After completing her PhD in Cultural Anthropology at Brown University in 2004, Alex commenced a diverse career at Intel Corporation. During her 15 years at Intel, Alex made significant contributions to the R&D and commercial development of technology across the Advanced Research, Digital Home and Internet of Things divisions, being listed as co-inventor on 11 patents. In 2016, Alex was appointed Principal Engineer in Social Science within the Internet of Things Division and was Intel's foremost domain expert and research practitioner in homes and home life. Alex has also authored multiple publications, presented at academic, regional and industry conferences and is a member of the American Anthropological Association and the Ethnographic Praxis in Industry Community.

Currently, she is interested in the role that social scientists play in exercising ethics, accountability, and data rights in the deployment of commercial cyber-physical systems.

Dr Jessica Herrington, Futures Specialist, School of Cybernetics

Jess Herrington integrates her expertise in neuroscience, creativity, and technology in her role as Futures Specialist at the ANU School of Cybernetics. Focusing on artificial intelligence, immersive technologies, and human perception, Jess is dedicated to redefining our interaction with and comprehension of emerging technologies.

Jess holds a PhD in neuroscience, specializing in human visual perception, along with degrees in Philosophy, Psychology, and Visual Art. Her professional journey before ANU involved collaborating with leading global brands to develop innovative digital experiences. Her projects span from creating AR/VR environments for film-makers to developing educational applications, notably an augmented reality app designed to enhance neuroanatomy education at the Australian National University Medical School.

Her artistic work has been featured at international conferences, art galleries, and festivals. As a committed advocate for inclusivity in technology, Jess has mentored young digital creators and spoken at institutions such as the Royal College of Arts in London and the Powerhouse Museum in Sydney. Her written work, exploring the intersections of art, technology and futures, appears in publications such as the Springer Series for Cultural Computing.

Safiya Okai-Ugbaje, Researcher and Lecturer, School of Cybernetics

Safiya Okai-Ugbaje completed her PhD at the University of Notre Dame Australia focusing on innovative solutions in the educational technology domain. Safiya holds a Bachelor of Technology in Mathematics and Computer Science from the Federal University of Technology Minna, Nigeria and a Master of Science in IT Management with Distinction from Staffordshire University, UK.

Safiya has worked on a variety of projects in the areas of cloud computing and green ICT, educational technologies, and cyber safety for Australia's Indigenous children. She is highly passionate about conducting research that moves beyond theory into practical applications that provide sustainable solutions to contextual problems. During her PhD, Safiya worked with various stakeholder groups including senior management and academics to facilitate the delivery of innovative and suitable educational solutions in higher education. The project succeeded in demonstrating how local opportunities may be effectively utilised to facilitate technology-enhanced teaching and learning even where resources are limited.

Thomas Biedermann, Senior Lecturer, School of Cybernetics

Thomas researches and develops educational methods to create safe, sustainable, and responsible systems. His approaches draw on cybernetics, visual arts, and playcentric and embodied methods, to support diverse participants in imagining and acting towards hopeful futures.

Thomas has applied this approach in a range of contexts, including a Sino-French joint research lab, the science diplomacy sections of French embassies in the USA and Australia, and academia. He has served as inaugural manager of the 3AI Institute at ANU, and program manager for the ANU-wide decarbonisation program, Below Zero.

Thomas enjoys combining analysis and his systems engineering training with the creativity and joyfulness of visual arts and gameful design.

Members to join at a later stage

Tarek Hamel, distinguished professor at the University of Côte d'Azur

Tarek joined the University of Côte d'Azur in 2003. He received his Ph.D. in Robotics from the University of Technology of Compiègne (UTC), France 1996. After two years as a research assistant at the UTC, he joined the Centre d'Etudes de Mécanique d'île de France in 1997 as an associate professor.

His research interests include nonlinear control theory, estimation, and vision-based control with applications to unmanned Aerial Vehicles.

He is a senior member of the Institut Universitaire de France (IUF) since 2017 (renewed in 2022) and IEEE fellow member since 2019. He is an editor of the Unmanned Systems Journal and associate editor. He has served in the past as an associate editor for IEEE Transactions on Control Systems Technologies, IEEE Transactions on Robotics, and Control Engineering Practice Journal. He is also an expert in evaluating the FP7, H2020, and Horizon Europe Research and Innovation programs of the European Commission. During his career, his publications have received a total of more than 14060 citations (data from Google Scholar), with an h-index of 53 and an i-index of 155.

ENSTA - Integration Proposal

IRL 2010 - CROSSING

ENSTA proposal to join 2026-2030

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Abstract

As part of the integration of ENSTA as a partner of the IRL Crossing, this document summarises the context and topics addressed in the laboratory. The second part highlights ENSTA's involvement in the laboratory's renewal.

Keywords: IRL CROSSING, ENSTA, Australia

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■ Context

Innovations in cyber-physical human systems (CPHS) require a multidisciplinary strategy, using knowledge in control theory, human-machine interaction, formal methods, autonomous systems, human factors, cognitive psychology, economics, and related fields. The Crossing Lab is dedicated to exploring interdisciplinary dimensions and crafting innovative technological solutions. This initiative brings together scholars and professionals to exchange insights into scientific and technological advancements and to deepen understanding of cyber-physical human systems. Research endeavours focus on the modelling, design, analysis, control, verification, and certification of CPHS, encompassing theoretical, algorithmic, computational, and experimental aspects. Key areas of focus include:

- Modelling, analysis, and control of integrated CPHS,
- Social and societal dimensions of CPHS,
- Security, privacy, and ethical considerations in CPHS,
- Human behaviour and collaboration between humans and autonomous systems,
- Ensuring safety and resilience in critical CPHS scenarios,
- Diverse applications of CPHS such as healthcare, transportation, human-space technology, and smart infrastructure.

Pioneering advances in CPHS require an interdisciplinary approach, drawing on expertise in control theory, human-machine interaction, formal methods, autonomous systems, human factors, cognitive psychology, economics, and related fields. The primary focus lies on the utilisation of AI-based algorithms.

This context was the settlement for the IRL Crossing Lab; after several years of activity, the next step for the IRL is to broaden the research spectrum. On top of this, the CNRS would like to merge the actual Crossing lab and the IRP ARS. The next sections describe the 2 labs.

■ Crossing and ARS projects

Crossing Partners

Crossing is partnered with CNRS, University of Adelaide, IMT Atlantique, University of South Australia, Flinders University, Naval Group. Crossing is a French Australian Laboratory for Human-Autonomous Agents Teaming. As examples, we can cite these Crossing projects:

- Fatigue and Team Performance in Maritime Environment
- Perceiving Distant Collaborative Activity with Mixed Reality
- Learning by Robot Manipulation Demonstration via GANs
- Adaptive Control of Underwater Robots with Machine Learning
- Human-Robot Interaction: Detect User Implicit Need of Help and Generate a Set of Assistive Actions

- Real-Time Human Performance Monitoring
- Predictive Maintenance With Experts In The Loop Solutions for Light Pollutions on Console Displays

Details are given here on the CNRS website : <https://crossing.cnrs.fr/>

IRP ARS

The Franco-Australian IRP-ARS is dedicated to Advanced Autonomy for Robotic Systems. It is managed by Professor Tarek HAMEL (I3S UMR-CNRS 7271, Cote d'Azur University) in collaboration with Professor Robert MAHONY (Australian Centre for Robotic Vision, Australian National University) and started in 2020.

Details are given here on the CNRS website : <https://melbourne.office.cnrs.fr/project/irp-ars/>

■ ENSTA's Involvement until 2024

ENSTA has participated in collaborative initiatives with South Australian universities, particularly in shaping the Autonomous Marine System chapter of Crossing Lab since 2018. The implications of ENSTA can be synthesised via several actions.

Staff Mobility:

- Visiting Professorship: One-year Visiting Professor in 2019/2020.
- Researcher at Crossing: 2 years of research engagement at Crossing from 2022 to 2024.
- Invited Professors: 2 invited professors from Flinders University, one in 2022 and another in 2024.

Joint Engagement via Students:

- Cotutelle Agreement: Collaboration through a cotutelle agreement with Flinders University.
- PhD Supervision: Supervision of 4 PhD students since 2019, with Thomas Chaffre being the first PhD candidate from the Lab.
- Internship Programs: Hosting 1-2 Baudin interns annually since program inception, hosting 2-3 master's students per year in Australia.
- Participation in PhD Panels: involvement in several PhD panels as reviewers.

Joint Projects:

- Cooperation with Naval Group Pacific: Collaboration led to an application to ARC Linkage and the supervision of one PhD candidates.

- Cooperation with UniSA: Collaboration for research on autonomous houseboats and human factors (CRC iMove tentative).
- Joint Project with Australian Partners: Collaboration on COLREGs.

Research Outputs:

- Joint publications (see bibliography) between French and Australians partners: a total of 16 joint publications (including 1 journal paper and 1 conference paper under review forthcoming in 2024) involving the four South Australian universities, contributed by Gilles Le Chenadec and Benoit Clement from ENSTA. *Note that only the post 2021-ones are identified in Crossing.*
- joint PhDs: 3 on going and 1 completed,
- Seminars: ENSTA delivers 3-4 seminars annually at Australian universities.

■ ENSTA participation as a stakeholder

Based on the ENSTA competences, the following topics can be part of the scientific project of Crossing Lab.

Perception

This research focuses on robotics and, more particularly, on perception and learning problems. The aim is to develop methods to simplify and make the use of robots more robust and to increase their autonomy. This can be done with the following research questions:

- navigation, mapping, localisation, path planning,
- learning applied to multi-modal perception and reinforcement learning,
- applications to mobile robots, UAVs and autonomous vehicles.

Control and guidance

This project focusses on mixing control / guidance techniques with AI for autonomous systems, particularly within maritime applications.

- Adaptive control enhanced with reinforcement learning. This consists of developing a simplified methodology for synthesising control laws for AUVs and maritime vessels, ensuring mathematical guarantees on robustness, reliability, safety, and performance; integration of reinforcement learning into adaptive control; and implementation of fault-tolerant control mechanisms based on reinforcement learning.
- Obstacle avoidance under a set of rules. This consists of novel strategies for rendezvous in cooperative or non-cooperative modes; Emphasis on human readability of control strategies; design of multi-constraint optimisation-based planners capable of identifying long-term, COLREG-compliant trajectories with a high navigation success rate; use of AIS history to predict vessel behaviour, akin to a navigator's experience, facilitating the development of methods to emulate mental models for long-term trajectory predictions.
- Multi-agent path planning accelerated by heuristics with exploration of learning from operational teaming with unmanned systems.

Interactions

This research aims to design a robotic system with social abilities to assist people with physical and cognitive impairments, with the aim of improving their quality of life. The project involves a robotic agent that supervises, encourages, and engages users in tasks such as rehabilitation, physical therapy, and cognitive stimulation. Innovative aspects include personalised therapy, continuous monitoring, and affordable sensing, tracking, and robotics technologies.

The associated research questions:

- Human-user modelling, including gestures and postures classification.
- Developing algorithms for automatic and incremental learning of concepts and tasks.
- Modelling and interpreting user physiological data.
- Signal processing for verbal and non-verbal communication in human-robot interaction.
- Developing learning algorithms and behaviour control architectures based on user preferences and disability levels.
- Understanding and incorporating social and cognitive sciences results for more realistic robotic solutions.

Industrial partners

As Naval Group is a stakeholder of Crossing Lab, the joint laboratory between Ensta and Naval Group will be complementary (as it happens with SENI and LC2IA). Other industrial partners happy to collaborate with Australian partners are welcome to participate.

Invited researchers

With two campus in Palaiseau and Brest, ENSTA can welcome Australian researchers, allowing them to meet all French partners in two regions.

People

At this stage, the possible involved permanent researchers are (alphabetical order):

1. Benoit CLEMENT,
2. David FILLIAT,
3. Gilles LE CHENADEC,
4. Bruno MONSUEZ,
5. Adriana TAPUS,
6. Christophe VIEL.

Resources

To join IRL CROSSING as a stakeholder ("tutelle"), ENSTA will have to contribute resources. This will need to be negotiated. As a suggestion, ENSTA offers to:

- Supervise at least 2 PhD students as already done with Flinders University. This comes with funding of two half-doctoral fellowships every three years.
- Provide offices spaces access for visiting academics in Palaiseau or Brest,
- Provide access to experimental facilities (robotics pool, robotics platforms and simulators)
- Affect 4 permanent members as secondary affiliation (including at least 0.5FTE Professor)
- the research contribution will be 30kEUR per year (can be partly in kind)

■ Contacts

ENSTA is represented:

- CEOs (Vice-Chancellor equivalent):
Elisabeth CREPON and Bruno GRUSELLE,
- Research Direction (DVCR equivalent):
Yann DOUTRELEAU and Laurent EL KAIM,
- Principal investigator:
Benoit CLEMENT.

■ Co-authored AU-FR Publications linked to Crossing

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1. Members Biographies

Prof. Benoit CLEMENT is a full professor at ENSTA Bretagne, France, while also serving as an Adjunct Professor at Flinders University. He currently spearheads the Information Science and Technology Department at ENSTA Bretagne, following his tenure as Deputy Head of the renowned Lab-STICC (CNRS UMR 6285) and his leadership of the Ocean Sensing and Mapping Department at ENSTA Bretagne. Presently situated in Adelaide, South Australia, Prof. Clement occupies the role of CNRS Researcher at CROSSING IRL CNRS 2010, where he engages in collaborative research endeavors with CNRS, Naval Group, Flinders University, University of Adelaide, and University of South Australia. His expertise and contributions extend beyond academia by his membership in IFAC technical committees on Robust Control and Marine Systems. Prof. Clement's academic journey is characterised by a commitment to excellence and scholarly inquiry. He earned his Master of Science degrees in Automatic Control and Signal Processing from University Paris-Saclay and in Automatic Control and Systems from CentraleSupélec in 1998. Subsequently, he pursued his Ph.D. at CentraleSupélec, awarded a Ph.D. degree in Physics from the University Paris-Saclay in 2001. In 2015, he furthered his academic credentials with the attainment of the "Habilitation à Diriger des Recherches" (HDR) degree in Automatic Control from the University of Brest. Currently, Prof. Clement's research endeavors focus on the integration of artificial intelligence with traditional approaches in maritime robotics Guidance, Navigation, and Control (GNC) algorithms. This research is supported by the French Defence Agency, the French Research Agency (CNRS), and collaborative efforts with industrial partners. Prof. Clement's dedication to advancing knowledge and innovation underscores his significant contributions to the field of control systems and maritime robotics.

David FILLIAT is professor in the Computer Science and System Engineering Laboratory (U2IS) at ENSTA Paris and the scientific director of the IP Paris Interdisciplinary Center for Defense and Security (CIEDS). His research focuses on robotics and more particularly on perception and learning problems. He is looking to develop methods to simplify and make more robust the use of robots and to increase their autonomy. He is particularly interested in:

- navigation, mapping, localization, path planning,
- learning applied to multi-modal perception and reinforcement learning,
- applications to mobile robots, UAVs and autonomous vehicles.

Gilles LE CHENADEC received his PhD in underwater acoustics from the Université de Bretagne Occidentale, France in 2004. He is currently an associate professor with ENSTA Bretagne (Brest, France) and a research fellow at Lab-STICC research laboratory. His research interests include topics related to signal processing and machine learning applied to AUV control and sonar data processing.

Adriana TAPUS is a Professor at ENSTA Paris, member of the Institut Polytechnique de Paris, in the Autonomous Systems and Robotics Laboratory of the Computer Science and Systems Engineering Unit (U2IS). In 2011, she obtained Habilitation (HDR) for her thesis titled "Towards a personalized Human-Robot Interaction". She obtained her PhD in Computer Science from the Swiss Federal Institute of Technology in Lausanne (EPFL), Switzerland in 2005 and her degree in Computer Engineering from Polytechnic University of Bucharest, Romania in 2001. She has worked as a Research Associate at the University of Southern California (USC), where she was among the pioneers in the development of socially assistive robotics, contributing mainly in machine learning, human modeling, and human-robot interaction. Prof. Tapus is Associate Editor of the International Journal on Social Robotics (IJSR), ACM Transactions on Human-Robot Interaction (THRI), and IEEE Transactions on Cognitive and Developmental Systems (TCDS) and is on the steering committee of several major robotics conferences (IROS, ICRA, HRI,

RO-MAN, etc.). She has more than 150 research publications and received the Romanian Academy Award for her contribution in assistive robotics in 2010. In 2016 she was nominated as one of the "25 women in robotics you need to know about". She has been the main coordinator of several national and international projects. Since 2019, she is also a founder member of the RoboticsByDesign lab. Since 2019, she is also the Director of the Doctoral School of IP Paris.

Bruno MONSUEZ graduated in 1989 from Ecole Polytechnique, and he is currently Professor at ENSTA Paris. He received his PhD in Computer Science from the Ecole Polytechnique in 1994. He used to be the Director of the Computer Science and System Engineering Department at ENSTA Paris. His current research interests are focused on developing and enhancing hierarchical compositional mathematical models that can be used to represent hardware and software components of complex embedded systems as well as formal verification techniques that allow a conjoint verification of functional and non-functional properties of the software as well as of the hardware on which the software is expected to run.

Christophe VIEL received the master's degree in advanced system and robotic and the Engineering degree from Arts et Metier ParisTech, Paris, France, in 2014, and the Ph.D. degree from ONERA (The French Aerospace Lab) in 2017. His Ph.D. topic was on determination of command control and observator for multi-agent system with reduced information by event-triggered. He holds a post-doctoral position with the University of Plymouth, U.K. His research topic includes controls and strategies for autonomous sailboat. He is currently researcher at Lab-STICC UMR CNRS 6285 at ENST Bretagne.

Future Director CV

Pr Cédric BUCHE

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CURRICULUM VITÆ

Professor - ENIB / R&D Director (CTO) Naval Group Pacific

HIGHER EDUCATION

2012	Habilitation ¹ in Computer Science, Université de Bretagne Occidentale (UBO), France
2005	PhD in Computer Science, UBO, France
2002	MSc in Computer Science, Université Rennes I, France
2002	Engineer , National Engineering School (ENIB), France

EXPERIENCE

2023 - present	R&D Director (CTO) , Naval Group Pacific
2018 - present	Full Professor , ENIB
2021 - 2023	CNRS delegation , IRL 2010, Australia
2018 - 2020	Professor (by courtesy), University of Miami.
2007 - 2018	Associate professor , ENIB.
2006 - 2007	Assistant associate professor, UBO
2005 - 2006	Assistant associate professor, ENIB

RESEARCH

Topics :	Artificial Intelligence, Virtual Reality, Machine Learning, Human Factors, Robotic
Applications :	Video games, Virtual Environment for Learning, Maritime
Supervisions :	19 PhD + 31 MSc + 6 engineers + 3 Post-doc

TEACHING

Main :	Master 2 program leader during 10 years (computer science)
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COMPETITIONS / AWARDS

2023	1st RoboCup@Home ²
2022	1st RoboCup@Home
2022	Elected " French of the year in Australia "
2021	3rd RoboCup@Home
2020	1st RoboCup@Home Education
2017	Finalist Angry Birds AI - IJCAI 2017
2016	Finalist Angry Birds AI - IJCAI 2016
2015	1st Angry Birds AI - IJCAI 2015
2014	Semi-Finalist Angry Birds AI - ECAI 2014
2013	Finalist Angry Birds AI - IJCAI 2013
2013	1st Man vs Machine Challenge - IJCAI 2013

1. Highest degree in French academic education, awarded after passing the tenure process, qualifies for full professorship

2. == world champion in social robotics

RESPONSIBILITIES

2019 - 2023 :	Leader - Team "Robobreizh" (~ 10 researchers), Lab-STICC/CROSSING/LITIS
2019 - 2021 :	Leader - Team "RAMBO" (~ 15 researchers), Lab-STICC.
2017 - 2021 :	Leader - Program "Artificial Intelligence" (~ 50 researchers), Lab-STICC.
2013 - 2018 :	Editor-in-Chief of the journal "International Journal of Virtual Reality" (IJVR)

CONTRACTS

Institutional	EU, ANR, AID (defense), USA funding, FUI (≈ 500 K €+ PhD salaries)
Industrial	CIFRE, Expertise, Contract (≈ 1 M €+ PhD salaries) : ArianeGroup, Thales, Segula, Ericsson, Naval Group

PATENT

2021	"Infobuna" : software for disease detection and grading of coffee using photos (num IDDN.FR.001.170002.000.S.P.2021.000.2100)
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PUBLICATIONS

Overview :	<i>Inter. Journals</i>	<i>Nat. Journals</i>	<i>Inter. Conf.</i>	<i>Preprints</i>	<i>National Conf.</i>	Total
	28	4	73	5	5	115

Metrics : **h-index : 21**
citations : 1600

Major Journals :

Q1 :	Neau M., Santos P., A.G. Bossier, Macvicar A. and C. Buche (2025). Mining Informativeness in Scene Graphs : Prioritizing Informative Relations in Scene Graph Generation for Enhanced Performance in Application. <i>Pattern Recognition Letters</i> . Accepted
Q1 :	Delamarre A., Shernoff E., Buche C., Frazier S., Gabbard J. and Lisetti C. (2021). The Interactive Virtual Training for Teachers (IVT-T) to Practice Classroom Behavior Management. <i>International Journal of Human - Computer Studies</i> . 152.
Q2 :	Habib Y., Papadakis P., Le Barz C., Fagette A., Gonçalves T. (2024). Real-time, Dense UAV Mapping by Leveraging Monocular Depth Prediction with Monocular-Inertial SLAM. <i>Advanced Robotics</i> , 38(21) pages 1555-1566

Major Conferences :

Rank A :	Cazorla R., Pionel L., Papadakis P. and Buche C. (2021). Bottleneck Identification to Semantic Segmentation of Industrial 3D Point Cloud Scene via Deep Learning. <i>International Joint Conference on Artificial Intelligence (IJCAI)</i> . pages 4877-4878
Rank A :	Delamarre, A., Buche C. and Lisetti C. (2021). Modeling Emotions for Training in Immersive Simulations (METIS) : a Cross-Platform Virtual Classroom Study. <i>IEEE International Symposium on Mixed and Augmented Reality (ISMAR)</i> . pages 78-83

Jury

Habilitation :	Reviewer (1), Member (1)
PhD :	Reviewer (7), President (4), Member (15)

Future Director Application Letter

Cover letter - Director position - IRL CROSSING 2010

Prof. Cédric BUCHE

Email: cedric.buche@cnrs.fr

January 8, 2025

To whom it may concern,

I am thrilled to submit my application for the position of director at CROSSING.

— *Background* —

First, let me introduce myself. I was awarded a PhD in **Artificial Intelligence (AI) and Virtual Reality (VR)** in 2005. I started a position as a lecturer at French National Engineering School (ENIB) in 2006 and became a member of the Lab-STICC (UMR CNRS). I defended my *Habilitation à Diriger des Recherches* (HDR) in 2012 and received qualification for a professorship (Informatic section) in 2013. I have been appointed a position as a Professor at ENIB in 2018. Within a major in AI and VR, I collaborated with cognitive psychologists as the relation between **Human and AI** is the core of my research.

Throughout my career, I have been honored with national award bonuses (PEDR 2011-2015, PES 2015-2019, and 2021-2025). Currently, I am supervising 5 PhD students (in progress) and have supervised the completion of 14 PhDs and 31 research master's degrees.

Between 2017 and 2021, I led the "artificial intelligence" program at Lab-STICC, which counted 50 permanent researchers. From 2020 to 2021, I led the RAMBO team at Lab-STICC (10 permanent researchers) focusing my research on **interactive robotics**.

In terms of international recognition, I served as a Professor (by courtesy) at the University of Miami from 2018 to 2020, and as a visiting Professor at Florida International University from 2016 to 2017. Currently, I hold positions as an adjunct Professor at the University of Adelaide and as a visiting Professor at Flinders University.

In France, I have been involved in several academic projects (ANR, FIU, EU) and have led numerous industrial projects, including 7 CIFRE and CIFRE-Defense. In terms of influence, I served as the **chief editor of the journal IJVR** (International Journal of Virtual Reality) for 5 years (2013-2018). This journal has been ranked "C" by ERA CORE.

Regarding teaching, I have an average annual workload of 250 hours, focused on computer science education at ENIB and research programs (i.e. master2). I had the privilege of **leading the master's program in computer science for 10 years** at ENIB.

In addition, I have been involved in the management of my institution by being elected to the board of directors and the pedagogical council. I have served as a reviewer for 6 PhD dissertations and 1 HDR, and 18 as examiner or president. In 2019, I had the opportunity to be nominated as a French expert to produce a **report on the progress of AI in Russia**, commissioned by the French Government.

Between 2021 and 2023, I was on **delegation at the Australian IRL "CROSSING"**, the international CNRS laboratory bringing together the three universities of Adelaide, Naval Group, IMT and CNRS. In 2022, I received the **"French of the Year in Australia" award**, a significant achievement in my career. As a researcher, my priority has been to transfer the algorithms I have developed into practical applications. As a result, I have participated in competitions with positive outcomes, mainly in IJCAI and RoboCup, winning the RoboCupHome contests in 2022 and 2023 (i.e., becoming **world champions in social robotics**).

Since september 2023, I started a position as CTO at Naval Group Pacific (NGP) to **lead Naval Group's R&D activities in Australia**. This involves overseeing a research program in collaboration with local industrial and academic partners. In 2023, NGP initiated PhD programs in collaboration with 3 universities (Flinders, Univ. of Adelaide, UniSA) and Naval Group Singapore. My role is to coordinate local activities while liaising with Naval Group in France. NGP have 9 employees, including administrative staff, VIE, researchers, engineers, and PhD candidates. This position is a secondment for 2 years (2023-2025) from ENIB.

— Motivation —

I have been engaged with the CROSSING project since its inception. For me it all began in 2019 when, by chance during an Air France strike, I found myself stranded in India with François Marie DUTHOIT, international dean of Naval Group. Subsequently, under the guidance of Gilles COPPIN, I actively contributed to shaping the scientific framework of the IRL, marking my first visit to Adelaide in 2019. The project was initiated with themes that are perfectly aligned with my research interests. I could not miss out on this adventure, so I applied in 2020 for a delegation to a laboratory (CROSSING) that did not yet exist, a gamble that paid off as CNRS accepted my request.

Thus, I relocated with my family to Adelaide for one year. We arrived in Australia with Jean-Philippe DIGUET in January 2021. My delegation position was renewed twice, which allowed me to build a lasting relationship of trust with all partners. In terms of outcomes, we produced joint publications, ongoing co-supervisions, and a twice in a row title of world champion in social robotics. Following these results, I was eager for new challenges while remaining driven in research. Joining Naval Group Pacific (NGP) as CTO allowed me to leverage my academic background and glean insights from the industry. A new momentum was launched in CROSSING by the company. A thesis program was initiated with CROSSING's academic partners, a NGP roadmap was defined, and a strong link with Naval Group Singapore and Naval Group France were established. A maritime platform is under development, a state-of-the-art tool that CROSSING will use to merge research outcomes.

The renewal project is indeed an exciting prospect.

CROSSING is strengthened by world class academic experts and holds a strategic stakeholder with the European leader in the maritime sector, Naval Group. The foundational work undertaken during CROSSING's nascent years has established a solid framework for a larger-scale project. As we have fostered mutual understanding, initiated collaborations, and tackled scientific challenges, the outcomes achieved within just three years are commendable. However, we must ramp up our efforts, and this implies **defining a strong, ambitious, and unique common roadmap**. It is through this endeavor that we will achieve success. The proposal "CROSSING 2.0" is aligned with this objective.

CROSSING addresses themes related to interactions between Humans and AIs. My academic expertise in AI, VR, human factors, and interactive robotics equips me to effectively address these scientific challenges and engage with the stakeholders of CROSSING. Furthermore, my understanding of Naval Group's operations and expectations is a significant asset. Finally, I am familiar with all operational staff partners, and I have strong knowledge of the Australian ecosystem. Therefore I am confident that I am qualified and experienced to lead the next CROSSING phase.

In addition, new potential CROSSING partners are global leaders in their respective fields. ENSTA, with whom I am already working with, will bring visibility to Europe and strengthen existing academic partnerships in Australia. ENSTA already has a joint laboratory with Naval Group in France, this as an incredible opportunity to pool interactions, research and resources. ANU is a globally renowned entity. Beyond the incredible opportunity for research collaborations, CROSSING's visibility would be enhanced. Such new partners would offer significant advantages to strengthen funding applications in France, SA or VIC in Australia, inter-states and at the European level. This new and strong collective will propel CROSSING's current project forward.

— Conclusion —

As a conclusion, I am convinced that my career background and my professional skills place me in a privileged position to meet with requirements as director of CROSSING.

Should you require any further information, I would be more than happy to provide additional information.

Yours sincerely,

A handwritten signature in blue ink, appearing to read 'C. BUCHE', with a stylized flourish at the end.

Prof. Cedric BUCHE

Adelaide, South Australia

